Comparing Reversible and Non-Reversible MCMC for Bayesian Mixture Models

Benjamin Robinson

August 2026

Abstract

Ascolani and Zanella [1] recently introduced a non-reversible lifted Metropolis–Hastings algorithm [2,14] for the discrete space of allocation variables in finite Bayesian mixture models [9], and obtained scaling limits in the uninformative data case indicating that the algorithm explores the state space in $\mathcal{O}(n)$ iterations, whereas their reversible random-scan Gibbs sampler converges in $\mathcal{O}(n^2)$ iterations. This is a speed-up by a factor of n, the sample size. Their samplers both involve a Metropolis mechanism that proposes a single allocation at a time. Roberts and Rosenthal [32] recently compared reversible and non-reversible (lifted) Metropolis–Hastings chains in the context of temperature moves in simulated tempering. They found that non-reversibility is most advantageous when proposals are microscopic. In this report, we generalize Ascolani and Zanella's [1] algorithms to allow for macroscopic proposals, and we find that in the uninformative-data case, when both our reversible and non-reversible algorithms are each optimally tuned, the advantage of non-reversibility is only moderate.

1 Introduction

Mixture models are a useful framework for representing heterogeneous data. Rather than assuming each observation is generated by a single mechanism, a mixture model allows us to represent the population as a collection of subpopulations, each described by its own distribution. This makes mixture models a flexible tool for clustering and density estimation [12, 22]. By combining simple component distributions, they can represent multimodality, varying scales, and other non-standard features of the underlying distribution [20, 22].

A Bayesian formulation of a mixture model is useful because it allows us to quantify uncertainty in the unknown component parameters, $\boldsymbol{\theta}$, and mixture weights, $\boldsymbol{w}$ [12, 20]. Mixture model likelihoods are often highly multimodal and can admit different parameter configurations that provide plausible explanations of the data [5]. A single point estimate can obscure this uncertainty. A further advantage of the Bayesian framework is that uncertainty in the latent population structure can be represented explicitly. We augment the data by associating each observation Y_i with an unobserved allocation variable $c_i \in \{1, \dots, K\}$, indicating the component from which it was generated. These latent allocations can then be inferred jointly with the component parameters and mixture weights. The full posterior

distribution therefore quantifies uncertainty not only in $(\boldsymbol{\theta}, \boldsymbol{w})$, but also in the allocation of observations between components [12, 20].

Markov chain Monte Carlo is well suited to this setting because augmenting the model with allocation variables turns posterior simulation into a sequence of simpler conditional sampling problems. Conditional on c and the observed data $\boldsymbol{Y} = (Y_1, \ldots, Y_n)$, the likelihood separates by component, and with suitable priors the full conditional distribution of $(\boldsymbol{\theta}, \boldsymbol{w})$ can often be sampled directly [9, 12]. The allocations can then be sampled from their conditional distribution $\pi(c \mid \boldsymbol{Y}, \boldsymbol{\theta}, \boldsymbol{w})$. Alternatively, we can integrate out $\boldsymbol{\theta}$ and $\boldsymbol{w}$ and construct an MCMC algorithm that targets the marginal posterior $\pi(c \mid \boldsymbol{Y})$ directly before sampling from $\pi(\boldsymbol{\theta}, \boldsymbol{w} \mid c, \boldsymbol{Y})$ [1]. In this report, we focus on the latter.

MCMC algorithms targeting $\pi(c \mid \boldsymbol{Y})$ can be computationally inefficient. Mixture posteriors can contain well-separated regions of high posterior probability, between which MCMC algorithms may move only rarely [5]. And the efficiency of an MCMC algorithm can deteriorate as the sample size n increases [1, 16]. One approach to improving MCMC efficiency is to use non-reversible samplers. A common construction is to augment the state space with a direction variable, allowing the chain to persist in one direction rather than repeatedly retracing its path [6, 8, 14]. Ascolani and Zanella [1] recently developed a non-reversible sampler targeting the marginal posterior $\pi(c \mid \boldsymbol{Y})$. Their algorithm works by first sampling a pair of components k and k', and then persistently proposing to move observations from k to k', changing a single allocation at each iteration, until a Metropolis rejection reverses the direction. They obtained scaling limits indicating that the non-reversible sampler converges in $\mathcal{O}(n)$ iterations, whereas the reversible random-scan Gibbs sampler converges in $\mathcal{O}(n^2)$ iterations [1]. This suggests that non-reversibility can provide a speed-up by a factor of n, the sample size.

It is important to note that both their samplers move only a single allocation at a time. In their analysis, they focused on the case with uninformative data, and they found that with their microscopic proposal the Metropolis acceptance probability is $1 - \mathcal{O}(n^{-1})$ [1]. Their random-scan Gibbs sampler may be written as a Metropolis–Hastings algorithm with acceptance probability identically 1. This high-acceptance setting is particularly relevant in light of the recent results of Roberts and Rosenthal [32]. They compared reversible and non-reversible MCMC samplers in the context of inverse-temperature moves in simulated tempering [32]. They found that the efficiency ratio between their non-reversible and reversible samplers can be written in terms of the asymptotic Metropolis acceptance rate, which is in turn a function of the proposal scale, ℓ:

$$\frac{\mathrm{eff}_{\mathrm{NR}}}{\mathrm{eff}_{\mathrm{R}}} = \frac{1}{1 - \mathrm{acc}}.$$

This ratio diverges as acc approaches one. In their setting, high acceptance, which corresponds to microscopic proposals, is precisely where non-reversibility is most beneficial. This motivates us to generalize the non-reversible algorithm in [1]. In this report, we study what happens when the proposal size is instead allowed to remain macroscopic, proportional to n, and find that the $\mathcal{O}(n)$ speed-up is not necessarily attributable to non-reversibility; rather, it may be a consequence of the microscopic $\mathcal{O}(1)$ proposal scale.

Our main contributions in this report are as follows:

1. We introduce two new tunable multiple-allocation algorithms. The reversible kernel $P_{\mathrm{R},\ell}$ and its lifted non-reversible counterpart $P_{\mathrm{NR},\ell}$ are defined for an arbitrary positive target probability distribution $\pi(c)$ on $[K]^n$. The algorithms work by first sampling a pair of components, then drawing $h \sim \mathrm{Unif}\{1, \ldots, \ell\}$ and proposing to move h allocations between the selected components. The case $\ell = 1$ recovers the corresponding one-allocation kernels in [1]. We prove that both $P_{\mathrm{R},\ell}$ and $P_{\mathrm{NR},\ell}$ are π-invariant and that both kernels generate ergodic chains.

2. We identify the large-n limits of both algorithms at a macroscopic proposal scale. For a two-component model in the case with uninformative-data case, we set ℓ_n to be a fixed proportion of the sample size n. Using Karr's weak-convergence theorem for Markov chains [18], we prove weak convergence of the laws on path space to explicit non-degenerate discrete-time (lifted) Metropolis–Hastings chains $P_{\mathrm{R},\tilde{\ell},\infty}$ and $P_{\mathrm{NR},\tilde{\ell},\infty}$. Notably, neither limit requires a time rescaling. This suggests that with macroscopic proposals both the reversible and non-reversible algorithms explore the state space on an $\mathcal{O}(1)$ time scale, and the $\mathcal{O}(n^2)$-versus-$\mathcal{O}(n)$ comparison with microscopic proposals is caused not by non-reversibility but by the proposal scale.

3. We compare the algorithms numerically both at stationarity and during the transient phase. Specifically, we compare the limit chains $P_{\mathrm{R},\tilde{\ell},\infty}$ and $P_{\mathrm{NR},\tilde{\ell},\infty}$ at stationarity across many different proposal scales $\tilde{\ell}$. We find that both algorithms perform best at a moderate proposal scale, and that the non-reversible sampler is only moderately more efficient than the reversible sampler when both algorithms are optimally tuned. When both chains use very large proposals, their efficiencies are nearly identical. When both chains use very small proposals, the reversible sampler's efficiency deteriorates more dramatically that that of the non-reversible sampler (Figure 4).

   We performed the same experiment on the finite algorithms $P_{\mathrm{R},\ell}$ and $P_{\mathrm{NR},\ell}$ using a synthetic dataset of size $n = 1{,}000$ and targeting its posterior. We found results very

similar to those for the limit chains. The main difference was that, when both algorithms were optimally tuned, the advantage of non-reversibility was smaller (Figure 5).

We simulated the transient phases of $P_{\mathrm{R},\ell}$ and $P_{\mathrm{NR},\ell}$ targeting posteriors based on synthetic datasets at a small proposal scale $\ell = 1$ and a moderate proposal scale $\ell = 300$. With $\ell = 1$, the non-reversible chain is much faster than the reversible chain. This agrees with the results in [1]. With $\ell = 300$, both chains approach stationarity rapidly, and the advantage of non-reversibility is small. We also timed these simulations and found that although the simulations with $\ell = 300$ proposals require more computational effort, they still reach stationarity much more quickly in clock time than the $\ell = 1$ chains.

In the rest of this report, we first provide background on non-reversible MCMC, with a focus on lifted Metropolis–Hastings, optimal scaling theory, and finite Bayesian mixture models (Sections 2–4). In Section 5, we review the previous work of Ascolani and Zanella [1] on non-reversible MCMC for Bayesian mixture models. In Section 6, we introduce our multiple-allocation generalizations $P_{\mathrm{R},\ell}$ and $P_{\mathrm{NR},\ell}$. In Section 7, we specialize to $K = 2$ in the prior case and state our weak convergence result (Theorem 2). In Section 8, we report our simulation results, and in Section 9, we discuss conclusions, limitations, and directions for further research.

2 Non-Reversible MCMC

Non-reversible Markov chains are those that do not satisfy the detailed balance equations. That is, a Markov chain on a finite set E with transition kernel K and target distribution π is non-reversible if it does *not* satisfy

$$\pi(x)K(x,y) = \pi(y)K(y,x), \qquad \text{for all } x, y \in E. \tag{1}$$

And reversible chains are those that *do* satisfy the detailed balance equations. Many commonly used Markov chain Monte Carlo algorithms, such as the class of Metropolis–Hastings algorithms [15, 23], are designed to satisfy the detailed balance equations because the equations are a simple-to-verify sufficient condition for the target π being an invariant distribution. This definition of non-reversibility can be generalized to include continuous-time Markov processes on continuous state spaces. It has long been known that non-reversible Markov processes can converge to stationarity much more efficiently than comparable reversible processes [8, 14, 17]. Non-reversibility has proved useful in Monte Carlo applications, where efficient samplers for complex target distributions, such as big-data Bayesian posteriors, can save substantial computational effort [1, 33]. Many novel non-reversible samplers

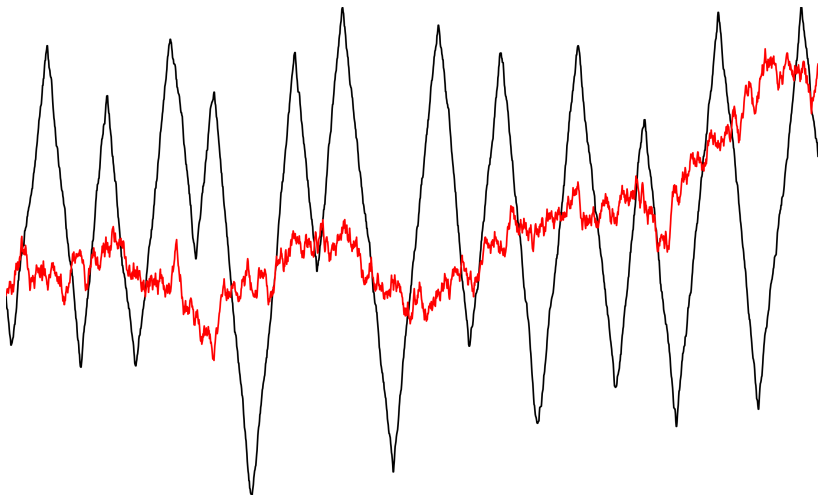


Figure 1: Illustrative trace plots of a reversible Metropolis–Hastings chain (red) and its non-reversible lifted counterpart (black), targeting the same invariant distribution. The reversible chain is diffusive and explores space slowly. The non-reversible chain persists in one direction for long periods of time and so covers a large part of the state space quickly. The acceptance rate of both chains here is ≈ 1.

have been developed in recent years [2–4], and in many cases such chains are substantially more efficient than their reversible counterparts [1, 37].

The intuition is that a reversible chain started from stationarity looks the same forwards and backwards in time, so it cannot maintain persistent motion in any particular direction. Consequently, reversible chains have a diffusive, backtracking character. A non-reversible chain can instead sustain momentum in a consistent direction. This can suppress backtracking and result in chains that sweep through the state space more efficiently. Figure 1 shows the effect. A well-known class of non-reversible algorithms is obtained by "lifting" reversible algorithms onto larger state spaces by introducing an auxiliary momentum variable [6, 8]. Consider the double-birth-death chain from [32]: a Markov chain on $\mathbb{Z} \times \{+, -\}$ with transition probabilities

$$\begin{aligned}
&\mathbb{P}\big((i,+),(i+1,+)\big) = A, \quad \mathbb{P}\big((i,+),(i-1,+)\big) = B, \quad \mathbb{P}\big((i,+),(i,-)\big) = C, \\
&\mathbb{P}\big((i,-),(i-1,-)\big) = A, \quad \mathbb{P}\big((i,-),(i+1,-)\big) = B, \quad \mathbb{P}\big((i,-),(i,+)\big) = C,
\end{aligned}$$

where $A + B + C = 1$ are non-negative constants with $C > 0$. If $C = 0$ and $A = B = 1/2$, then the chain is the reversible random walk on a single copy of $\mathbb{Z}$. In this case, by the central limit theorem, it takes $\mathcal{O}(n^2)$ steps for the chain to travel an $\mathcal{O}(n)$ distance. If instead $C = 1/n$, $A = 1 - 1/n$ and $B = 0$, then the chain travels a distance n in n steps with probability $(1 - 1/n)^n \approx e^{-1}$ (see [8, 32]). In this simple example, the non-reversible chain explores the state space much more efficiently. This speed-up is very much problem-

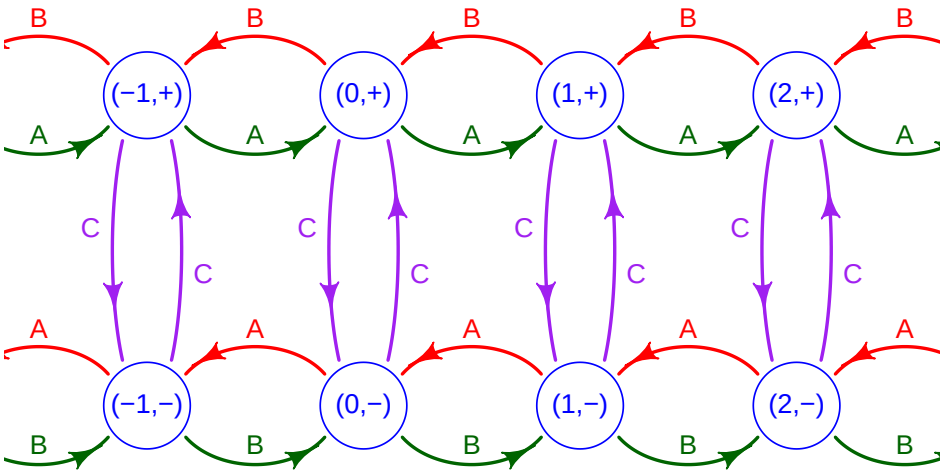


Figure 2: The lifted random walk on $\mathbb{Z}$. Diagram from [32].

dependent. In some cases, the speed-up provided by non-reversibility is only moderate [32]. In this report, we compare a reversible Metropolis–Hastings algorithm with a comparable non-reversible *lifted* Metropolis–Hastings algorithm targeting the posterior on the discrete space of allocation variables in the context of Bayesian mixture models.

3 Lifted Metropolis–Hastings and Optimal Scaling

Lifted Metropolis–Hastings algorithms [2,14] are a useful class of MCMC algorithms. For illustration, consider again the lifted space $\mathbb{Z} \times \{-1, +1\}$. Let π be an arbitrary probability measure on $\mathbb{Z}$ and consider the extended probability measure on $\mathbb{Z} \times \{-1, +1\}$ given by

$$\tilde{\pi}(i, v) = \frac{\pi(i)}{2},$$

whose marginal in i is π. Given two proposal kernels $\{q_+(i, \cdot)\}_{i \in \mathbb{Z}}$ and $\{q_-(i, \cdot)\}_{i \in \mathbb{Z}}$, let P_{lift} be the non-reversible $\tilde{\pi}$-invariant kernel defined by Algorithm 1.

Algorithm 1 Generating a sample $(i', v') \sim P_{\text{lift}}((i, v), \cdot)$

1: Sample $\tilde{i} \sim q_v(i, \cdot)$.
2: Set $(i', v') = (\tilde{i}, v)$ with probability

$$\min\left\{1, \frac{\pi(\tilde{i})q_{-v}(\tilde{i}, i)}{\pi(i)q_v(i, \tilde{i})}\right\}.$$

3: Otherwise, set $(i', v') = (i, -v)$.

This is simply a Metropolis–Hastings algorithm that flips the velocity v and switches to another proposal kernel after a rejection. Proposal kernels often have a spatial scale

parameter, which we call σ. For instance, we may take $q_{\pm}(i, \cdot) = \text{Unif}\{i \pm j : j \in [\sigma]\}$, where $[m] := \{1, \ldots, m\}$. Setting σ appropriately is important for designing efficient chains. Large σ gives ambitious proposals that are rarely accepted; small σ gives conservative proposals that are almost always accepted, and the chain crawls very slowly (see Figure 1). The optimal σ is problem-dependent and lies between these extremes. For a reasonable efficiency measure—such as the expected squared jumping distance $\text{eff}(\sigma) := \text{ESJD}(\sigma)$ or the effective sample size $\text{eff}(\sigma) := \text{ESS}_g(\sigma)$ for an observable g—efficiency typically deteriorates at both extremes.

Scaling is particularly important in high-dimensional problems because, for many high-dimensional targets, probability mass concentrates in relatively narrow regions [39]. In this context, Roberts, Gelman and Gilks [27] studied targets that are product i.i.d. distributions of the form $\pi_d(x_1, \ldots, x_d) = \prod_{i=1}^{d} f(x_i)$ and studied the limit as $d \to \infty$ of the first coordinate of a reversible Metropolis chain targeting π_d. Under some regularity conditions, they showed that obtaining a non-trivial limit requires accelerating time by d and scaling space by setting $\sigma_d = \ell/\sqrt{d}$. Under the same conditions, they identified the efficiency-maximizing value $\ell_{\text{opt}} > 0$, which depends on the Fisher information of f. Importantly, they also showed that efficiency may be written as a function of the acceptance rate $\text{acc}(\ell)$, and that $\text{acc}(\ell_{\text{opt}}) = 0.234$ to three decimal places for every suitable f. These results opened up the study of optimal scaling for MCMC. The ideas have since been extended to more general high-dimensional targets [29, 40], to discrete state spaces [26], and to large-sample Bayesian computation, where many posteriors concentrate as the number of observations $n \to \infty$ [33]. Even in settings where the theory of [29] does not obviously apply, an acceptance rate of 0.234 is often found to be at least near-optimal [10, 34]. Thus, even when the regularity conditions are restrictive, studying optimal scaling in this manner can reveal fundamental qualitative behaviour of algorithms and provide useful heuristics for tuning algorithms.

Roberts and Rosenthal [32] recently quantified the speed-up from non-reversibility in the context of simulated tempering [21]. They derived the efficiencies of reversible and non-reversible inverse-temperature processes as a function of the acceptance rate. Their efficiency measure is the speed of the limiting diffusion (see [31] for more detail). The two curves are shown in Figure 3. The optimal acceptance rate in the non-reversible case is 0.387, larger than the reversible optimum of 0.234. The efficiency at the non-reversible optimum exceeds that at the reversible optimum, but only by the ratio $1.89/1.33 \approx 1.42$: a 42% speed-up. They show [32] that the ratio of the efficiency of their non-reversible sampler to that of their reversible sampler, as a function of the acceptance rate, is

$$\frac{\text{eff}_{\text{NR}}(\text{acc}(\ell))}{\text{eff}_{\text{R}}(\text{acc}(\ell))} = \frac{1}{1 - \text{acc}(\ell)}. \tag{2}$$

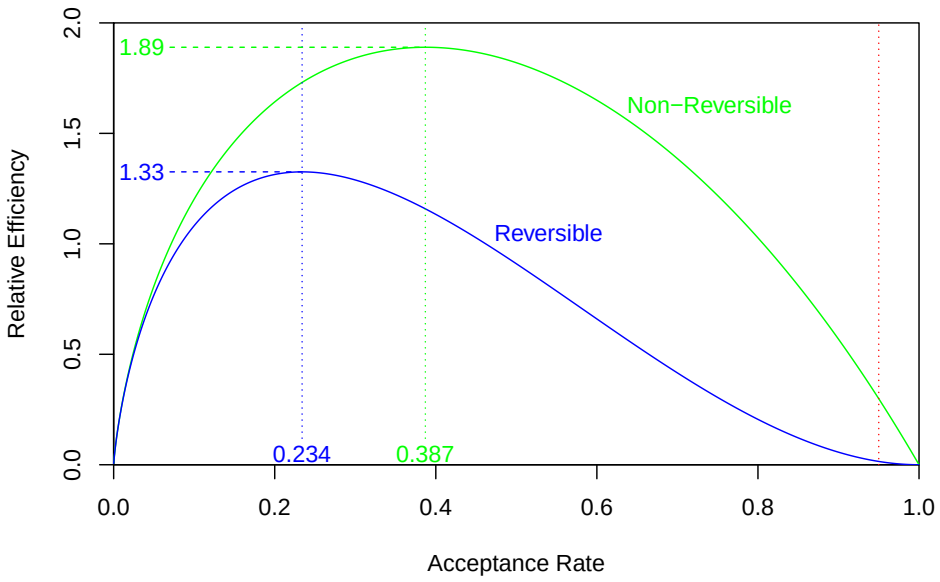


Figure 3: Efficiency curves for the non-reversible (top, green) and reversible (bottom, blue) tempering algorithms of [32], with their optimal scaling values (dashed lines). See [32] for details.

This efficiency ratio reveals a lot about the comparison of their algorithms. As ℓ increases, $\text{acc}(\ell)$ decreases and the non-reversible algorithm provides a negligible advantage over the reversible algorithm. As $\ell \to 0$, we have $\text{acc}(\ell) \to 1$ and the efficiency ratio diverges to ∞, which suggests that the non-reversible advantage grows as the acceptance probability approaches one. When ℓ is a moderate value between these extremes, both algorithms are most efficient, and the non-reversible algorithm provides only a moderate advantage. Although these results are specific to the simulated tempering setup in [32], we investigate whether reversible and non-reversible algorithms on the space of allocation variables in Bayesian mixture models exhibit similar qualitative behaviour.

4 Bayesian Mixture Models

A finite Bayesian mixture model with $K \geq 2$ components is given by

$$\begin{aligned} Y_i \mid \boldsymbol{\theta}, \boldsymbol{w} &\overset{\text{i.i.d.}}{\sim} \sum_{k=1}^{K} w_k f_{\theta_k}(\cdot), \quad i \in [n], \\ \theta_k &\overset{\text{i.i.d.}}{\sim} p_0, \qquad\qquad k \in [K], \\ \boldsymbol{w} &\sim \text{Dir}(\boldsymbol{\alpha}), \end{aligned} \tag{3}$$

where $\boldsymbol{\theta} = (\theta_1, \ldots, \theta_K)$, $\boldsymbol{w} = (w_1, \ldots, w_K)$, $\boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_K)$, and $\text{Dir}(\boldsymbol{\alpha})$ is the Dirichlet distribution on the $(K-1)$-dimensional simplex $\Delta_{K-1} \subset \mathbb{R}^K$. Data points Y_i lie in a space

$\mathcal{Y}$, and $f_\theta : \mathcal{Y} \to [0, \infty)$ is a probability density indexed by $\theta \in \Theta \subseteq \mathbb{R}^{d_\theta}$, with prior density p_0 assigned to θ.

Posterior inference with (3) is challenging. Even when f_θ belongs to an exponential family and p_0 is conjugate, the posterior $\pi(\boldsymbol{\theta}, \boldsymbol{w} \mid \boldsymbol{Y})$ is a mixture over the K^n possible allocations of the data to components [9], and explicit summation over these allocations becomes infeasible as n grows. A popular strategy is therefore to augment (3) with *allocation variables*, as in the classical treatments of incomplete-data problems [7]:

$$Y_i \mid c, \boldsymbol{\theta}, \boldsymbol{w} \overset{\text{i.i.d.}}{\sim} f_{\theta_{c_i}}(\cdot), \qquad c_i \mid \boldsymbol{\theta}, \boldsymbol{w} \overset{\text{i.i.d.}}{\sim} \mathrm{Cat}(\boldsymbol{w}), \qquad \boldsymbol{w} \sim \mathrm{Dir}(\boldsymbol{\alpha}), \qquad \theta_k \overset{\text{i.i.d.}}{\sim} p_0, \tag{4}$$

where $c = (c_1, \ldots, c_n) \in [K]^n$ and $\mathrm{Cat}(\boldsymbol{w})$ is the categorical distribution with weights $\boldsymbol{w}$. Given $\boldsymbol{Y} = (Y_1, \ldots, Y_n)$, the joint posterior of $(c, \boldsymbol{\theta}, \boldsymbol{w})$ is

$$\begin{aligned} \pi(c, \boldsymbol{\theta}, \boldsymbol{w} \mid \boldsymbol{Y}) &\propto \left[\prod_{i=1}^{n} w_{c_i} f_{\theta_{c_i}}(Y_i)\right] \mathrm{Dir}(\boldsymbol{w} \mid \boldsymbol{\alpha}) \prod_{k=1}^{K} p_0(\theta_k) \\ &\propto \prod_{k=1}^{K} \left[w_k^{n_k(c)+\alpha_k-1} \, p_0(\theta_k) \prod_{i:c_i=k} f_{\theta_k}(Y_i) \right], \end{aligned} \tag{5}$$

where $n_k(c) = \sum_{i=1}^{n} \mathbb{1}(c_i = k)$ and $\mathbb{1}$ denotes the indicator function. Integrating $(\boldsymbol{\theta}, \boldsymbol{w})$ out of (5) gives the marginal posterior of c,

$$\pi(c \mid \boldsymbol{Y}) \propto \left[\prod_{k=1}^{K} \Gamma(\alpha_k + n_k(c))\right] \prod_{k=1}^{K} \int_\Theta \prod_{i:c_i=k} f_{\theta_k}(Y_i) p_0(\theta_k) \mathrm{d}\theta_k. \tag{6}$$

If we can sample from $\pi(c)$, where from now on we suppress the dependence on the data for brevity, then we can sample from $\pi(\boldsymbol{\theta}, \boldsymbol{w} \mid c)$ and so perform full posterior inference. We therefore focus on sampling c. The computational challenge for mixture models shifts from evaluating a large number of terms to constructing Markov chains that explore $[K]^n$ efficiently.

5 Non-Reversible MCMC for $\pi(c)$

5.1 A Reversible Kernel and Its Lifting

Ascolani and Zanella [1] developed a non-reversible sampler targeting $\pi(c)$ with Markov kernel P_{NR}. It is a lifted Metropolis–Hastings algorithm. The reversible algorithm that they compare with is a random-scan Gibbs sampler given by

$$P_{\mathrm{MG}}(c, c') = \frac{1}{n} \sum_{i=1}^{n} \mathbb{1}\big(c'_{-i} = c_{-i}\big) \, \pi\big(c'_i \mid c_{-i}\big), \qquad c, c' \in [K]^n, \tag{7}$$

which updates a uniformly chosen allocation from its full conditional and is π-reversible. One of the key ideas behind P_{NR} is the choice of directions.

Choosing directions along which a non-reversible chain can persist is non-trivial and problem-dependent. It is not obvious how to do so in a way that produces long excursions. One could rely on the product structure of $[K]^n$ and attach a velocity component to each data point, which is what the lifting construction applied to the random-scan Gibbs kernel P_{MG} produces. Such directions are poorly suited to the geometry of $\pi(c)$ in (6). Instead, the geometry lends itself to efficient exploration when the persistent momentum is oriented along *pairs of clusters* [1].

For the moment, let π be an arbitrary probability measure on $[K]^n$, and write

$$\mathcal{K} = \left\{(k, k') \in [K]^2 : k < k'\right\} \tag{8}$$

for the set of ordered component pairs. As an intermediate step towards P_{NR}, Ascolani and Zanella first consider the π-reversible kernel on $[K]^n$ defined by

$$P_{\mathrm{R}}(c, c') = \sum_{(k,k') \in \mathcal{K}} p_c(k, k') P_{k,k'}(c, c'), \qquad c, c' \in [K]^n, \tag{9}$$

where

$$p_c(k, k') = \frac{n_k(c) + n_{k'}(c)}{(K-1)n}, \qquad (k, k') \in \mathcal{K}. \tag{10}$$

Here p_c is a probability measure on $\mathcal{K}$ for each $c \in [K]^n$, and $P_{k,k'}$ is the π-reversible Metropolis–Hastings kernel that proposes moving a uniformly selected allocation from cluster k to cluster k', or vice versa, each with probability $1/2$. The kernel is given explicitly by Algorithm 2.

Algorithm 2 Generating a sample $c' \sim P_{\mathrm{R}}(c, \cdot)$

Sample $(k, k') \sim p_c$.
Set $(k_-, k_+) = (k, k')$ with probability $1/2$ and $(k_-, k_+) = (k', k)$ otherwise.
if $n_{k_-}(c) < 1$ **then**
    Set $c' = c$.
else
    Sample $i \sim \mathrm{Unif}\left(\{i' \in [n] : c_{i'} = k_-\}\right)$, and set the candidate $\tilde{c} = (c_{-i}, k_+)$.
    Set $c' = \tilde{c}$ with probability $\min\{1, r(c, i, k_-, k_+)\}$, and set $c' = c$ otherwise, where

$$r(c, i, k_-, k_+) = \left(\frac{n_{k_-}(c)}{n_{k_+}(c) + 1}\right) \frac{\pi(\tilde{c})}{\pi(c)}. \tag{11}$$

end if

The lifted target is

$$\widetilde{\pi}(c, v) := \pi(c) \left(\tfrac{1}{2}\right)^{K(K-1)/2}, \qquad c \in [K]^n, \quad v = (v_{k,k'})_{(k,k')\in\mathcal{K}} \in \{-1, +1\}^{K(K-1)/2}, \tag{12}$$

and the $\tilde{\pi}$-invariant kernel P_{NR} is

$$P_{\mathrm{NR}}\big((c, v), (c', v')\big) = \sum_{(k,k')\in\mathcal{K}} p_c(k, k')\, \tilde{P}_{k,k'}\big((c, v), (c', v')\big), \tag{13}$$

where $\widetilde{P}_{k,k'}$ acts on the kth and k'th clusters and is the lifted counterpart of $P_{k,k'}$. Like (9), the kernel is a mixture over the $K(K-1)/2$ *axes* (k, k'), and each axis carries its own velocity component $v_{k,k'} \in \{-1, +1\}$, which fixes the *direction* of the proposal on that axis. $v_{k,k'} = +1$ proposes moving an allocation from k to k', and $v_{k,k'} = -1$ from k' to k. Only the component on the chosen axis is used, and possibly flipped, in a given iteration.

Algorithm 3 Generating a sample $(c', v') \sim P_{\mathrm{NR}}((c, v), \cdot)$

1: Sample $(k, k') \sim p_c$.
2: With probability ξ/n, flip $v_{k,k'}$ to $-v_{k,k'}$.
3: Set $(k_-, k_+) = (k, k')$ if $v_{k,k'} = +1$, and $(k_-, k_+) = (k', k)$ if $v_{k,k'} = -1$.
4: **if** $n_{k_-}(c) = 0$ **then**
5: Set $(c', v') = (c, v^{\mathrm{flip},(k,k')})$.
6: **else**
7: Sample $i \sim \mathrm{Unif}\left(\{i' \in [n] : c_{i'} = k_-\}\right)$.
8: Set $(c', v') = \big((c_{-i}, k_+), v\big)$ with probability $\min\{1, r(c, i, k_-, k_+)\}$, and otherwise set $(c', v') = (c, v^{\mathrm{flip},(k,k')})$.
9: **end if**
10: With probability ξ/n, flip $v'_{k,k'}$ to $-v'_{k,k'}$.

Here, $v^{\mathrm{flip},(k,k')} = \big(v_{-(k,k')}, -v_{k,k'}\big)$ denotes the velocity vector that agrees with v except that the (k, k') component is negated. The refreshment rate $\xi \in (0, n)$ ensures aperiodicity. Numerically, Ascolani and Zanella showed that P_{NR} outperforms P_{MG}, especially in the transient phase, and their scaling-limit analysis suggests that the improvement is by a factor of n.

5.2 Scaling Limits in the Prior Case

Ascolani and Zanella study the scaling limits of P_{MG} and P_{NR} targeting the prior $\pi(c)$. Formally, this is the case $f_\theta(y) = f(y)$, so that the likelihood carries no information about the labels.

Let

$$X_k(c) = \frac{n_k(c)}{n} = \frac{1}{n}\sum_{i=1}^{n} \mathbb{1}(c_i = k), \qquad c \in [K]^n,$$

be the proportion of data allocated to component k, and write

$$\mathbf{X}_t = \left(X_1(c^{(t)}), \ldots, X_K(c^{(t)})\right), \qquad \mathbf{V}_t = \left(v_{k,k'}^{(t)}\right)_{(k,k')\in\mathcal{K}}, \tag{14}$$

for the proportion vector and velocity components of the chain at time t. With an uninformative likelihood, the dynamics of P_{NR} and P_{MG} depend on c only through the cluster counts, so $\{\mathbf{X}_t\}$ under P_{MG} and $\{(\mathbf{X}_t, \mathbf{V}_t)\}$ under P_{NR} are themselves Markov chains [1].

Following [27,29], Ascolani and Zanella study the time-changed processes $Z_t^{(n)} = \mathbf{X}^{(n)}_{\lceil \tau(n)t \rceil}$ with $\tau(n) \to \infty$. If $\{Z_t^{(n)}\}_{t\geq 0}$ converges in a suitable sense to a non-degenerate ergodic process $\{Z_t\}_{t\geq 0}$, this suggests that $\mathcal{O}(\tau(n))$ iterations of the discrete-time chain are needed to mix [1]. They show that for P_{MG} the required time change is $\tau(n) = n^2$, with a Wright–Fisher diffusion in the limit, and that for P_{NR} it is $\tau(n) = n$, with a piecewise-deterministic Markov process in the limit.

In the prior case, the Metropolis–Hastings acceptance probability of P_{NR} is $1-\mathcal{O}(n^{-1})$ [1]. The random-scan Gibbs sampler P_{MG} can be written as a Metropolis–Hastings algorithm with acceptance probability identically one. Thus, the comparison above is made in the setting where acc $\to 1$. In the tempering setting of Section 3, this is precisely where non-reversibility is most beneficial, but not where the non-reversible algorithm is well tuned.

6 Multiple-Allocation Samplers

Our algorithms generalize P_{R} and P_{NR} by introducing a tunable scale $\ell \in \mathbb{N}$. In the original kernels, each proposal moves exactly one allocation between the chosen pair of clusters. We instead draw a step size $h \sim \mathrm{Unif}([\ell])$ and propose moving h allocations at once from k_- to k_+, so that $\ell = 1$ recovers P_{R} and P_{NR}. The reversible version is Algorithm 4, and the non-reversible lifted version is Algorithm 5. Throughout, $c^{\mathcal{I}}$ denotes the allocation vector obtained from c by reassigning every index in $\mathcal{I}$ to k_+.

Algorithm 4 Generating a sample $c' \sim P_{\mathrm{R},\ell}(c, \cdot)$

Sample $(k, k') \sim p_c$.
Set $(k_-, k_+) = (k, k')$ with probability $1/2$, and $(k_-, k_+) = (k', k)$ otherwise.
Sample $h \sim \mathrm{Unif}([\ell])$.
if $n_{k_-}(c) < h$ **then**
    Set $c' = c$.
else
    Sample

$$\mathcal{I} \sim \mathrm{Unif}\left(\{\mathcal{I}' \subseteq [n] : |\mathcal{I}'| = h,\ c_i = k_- \text{ for every } i \in \mathcal{I}'\}\right).$$

    Set $c' = c^{\mathcal{I}}$ with probability $\min\{1, r_h(c, \mathcal{I}, k_-, k_+)\}$, and set $c' = c$ otherwise.
end if

Here

$$r_h(c, \mathcal{I}, k_-, k_+) = \frac{\binom{n_{k_-}(c)}{h}}{\binom{n_{k_+}(c)+h}{h}} \frac{\pi(c^{\mathcal{I}})}{\pi(c)}. \tag{15}$$

Algorithm 5 Generating a sample $(c', v') \sim P_{\mathrm{NR},\ell}((c, v), \cdot)$

Sample $(k, k') \sim p_c$.
With probability ξ/n, replace $v_{k,k'}$ by $-v_{k,k'}$.
Set $(k_-, k_+) = (k, k')$ if $v_{k,k'} = +1$, and $(k_-, k_+) = (k', k)$ if $v_{k,k'} = -1$.
Sample $h \sim \mathrm{Unif}([\ell])$.
if $n_{k_-}(c) < h$ **then**
    Set $(c', v') = (c, v^{\mathrm{flip},(k,k')})$.
else
    Sample

$$\mathcal{I} \sim \mathrm{Unif}\left(\{\mathcal{I}' \subseteq [n] : |\mathcal{I}'| = h,\ c_i = k_- \text{ for every } i \in \mathcal{I}'\}\right).$$

    Set $(c', v') = (c^{\mathcal{I}}, v)$ with probability $\min\{1, r_h(c, \mathcal{I}, k_-, k_+)\}$, and otherwise set $(c', v') = (c, v^{\mathrm{flip},(k,k')})$.
end if
With probability ξ/n, replace $v'_{k,k'}$ by $-v'_{k,k'}$.

Theorem 1. *Let π be a probability measure on $[K]^n$ with $\pi(c) > 0$ for every $c \in [K]^n$, let $\ell \in [n]$, and let $\xi \in (0, n)$. Then $P_{\mathrm{R},\ell}$ is π-reversible and $P_{\mathrm{NR},\ell}$ is $\tilde{\pi}$-invariant, and both generate ergodic Markov chains, with stationary distributions $\pi(c)$ and $\tilde{\pi}(c, v)$, respectively.*

The proof is in Appendix A.

7 Scaling Limits at Macroscopic Step Size

As in [1], we work in the prior case and on the proportion space. We restrict to $K = 2$. This case still demonstrates what we want to show. Generalizing to $K > 2$ is a natural direction for future work. The corresponding weak convergence proofs are very similar, but they require more technical details.

When $K = 2$, the pair set $\mathcal{K}$ is a singleton and (10) gives $p_c(1, 2) \equiv 1$, so no axis is selected and the velocity is a single sign $v^{(t)} := v^{(t)}_{1,2} \in \{-1, +1\}$. Writing $x = x_1$, so that $x_2 = 1 - x$, and setting $\alpha = \alpha_1$ and $\beta = \alpha_2$, we abbreviate

$$X_t := X_1\big(c^{(t)}\big) = \frac{n_1\big(c^{(t)}\big)}{n} \in \Big\{0, \tfrac{1}{n}, \ldots, 1\Big\}.$$

In the prior case, the pushforward of the marginal target (6) under X is the rescaled Beta-Binomial distribution.

$$\pi_n(x) \propto \frac{\Gamma(nx+\alpha)\Gamma\big(n(1-x)+\beta\big)}{\Gamma(nx+1)\Gamma\big(n(1-x)+1\big)}, \qquad x \in \Big\{0, \tfrac{1}{n}, \dots, \tfrac{n-1}{n}, 1\Big\}. \tag{16}$$

The proposal draws the moved indices uniformly from the donor cluster, so every transition probability of $P_{\mathrm{R},\ell}$ and $P_{\mathrm{NR},\ell}$ depends on the current state only through (X, v). It follows that the processes X and (X, v) are Markov. If the initial distribution of c_0 is exchangeable within the fibres of X_0, then, by [28, Corollary 2], we obtain equality of the following total variation distances:

$$\big\|\mathcal{L}(c^{(t)}) - \pi\big\|_{\mathrm{TV}} = \big\|\mathcal{L}(X_t) - \pi_n\big\|_{\mathrm{TV}}, \qquad \big\|\mathcal{L}(c^{(t)}, v^{(t)}) - \tilde{\pi}\big\|_{\mathrm{TV}} = \big\|\mathcal{L}(X_t, v^{(t)}) - \tilde{\pi}_n\big\|_{\mathrm{TV}}.$$

We now take the step size to be a fixed proportion of the sample size,

$$\ell_n = \tilde{\ell} n, \qquad \tilde{\ell} = a/b \in (0,1) \cap \mathbb{Q} \text{ fixed}, \tag{17}$$

along the subsequence of n for which $\tilde{\ell} n$ is an integer. In this setting, the limiting acceptance probability does not converge to 1, nor does it converge to 0. Writing $h = n\tilde{h}$, with $\tilde{h} \sim \mathrm{Unif}\{1/n, \dots, \ell_n/n\}$, the ratio $\frac{\pi(c')q(c' \to c)}{\pi(c)q(c \to c')}$ is the following product of eight Gamma functions:

$$r_n(x, \tilde{h}) = \frac{\Gamma\big(n(x-\tilde{h})+\alpha\big)\Gamma(nx+1)\Gamma\big(n(1-x+\tilde{h})+\beta\big)\Gamma\big(n(1-x)+1\big)}{\Gamma\big(n(x-\tilde{h})+1\big)\Gamma(nx+\alpha)\Gamma\big(n(1-x+\tilde{h})+1\big)\Gamma\big(n(1-x)+\beta\big)}. \tag{18}$$

It is shown in Appendix C (Lemma 10) that $\min\{1, r_n\}$ converges to the Metropolis–Hastings acceptance probability (19) for the Beta density. By contrast, when $\ell = 1$, $\tilde{h} = 1/n \to 0$ and $r_n \to 1$.

The limiting kernels are the natural continuum analogues, given by Algorithms 6 and 7. Here $m_{v,\infty}$ is the Metropolis–Hastings acceptance probability for the Beta(α, β) density π_∞ under a Unif$[0, \tilde{\ell}]$ proposal,

$$m_{v,\infty}(x, \tilde{h}) = \min\left\{1, \ \frac{\pi_\infty(x - v\tilde{h})}{\pi_\infty(x)}\right\}, \qquad v \in \{-1, +1\}, \tag{19}$$

so that $v = +1$ corresponds to a downward move $x \mapsto x - \tilde{h}$ and $v = -1$ to an upward move $x \mapsto x + \tilde{h}$. Ratios with the denominator $\pi_\infty(x) = 0$ are treated as $+\infty$, so a proposal from a zero-density endpoint is always accepted.

Theorem 2. *Suppose $K = 2$, $\alpha, \beta \geq 1$, and $\tilde{\ell} = a/b \in (0,1) \cap \mathbb{Q}$. Let $n_j = jb$, set $\ell_j = \tilde{\ell} n_j = ja$, and keep the refreshment parameter $\xi > 0$ fixed. Under these conditions, the laws of the reversible chains generated by P_{R,ℓ_j} converge weakly on path space to the law of the chain generated by $P_{\mathrm{R},\tilde{\ell},\infty}$. The analogous convergence holds for the non-reversible chains generated by P_{NR,ℓ_j} with limiting kernel $P_{\mathrm{NR},\tilde{\ell},\infty}$.*

Proof. See Appendix C. □

Algorithm 6 Generating a sample $x' \sim P_{\mathrm{R},\tilde{\ell},\infty}(x,\cdot)$

Sample $v \sim \mathrm{Unif}\{-1,+1\}$ and $\tilde{h} \sim \mathrm{Unif}[0,\tilde{\ell}]$.
if $x - v\tilde{h} \notin [0,1]$ **then**
    Set $x' = x$.
else
    Set $x' = x - v\tilde{h}$ with probability $m_{v,\infty}(x,\tilde{h})$, and $x' = x$ otherwise.
end if

Algorithm 7 Generating a sample $(x',v') \sim P_{\mathrm{NR},\tilde{\ell},\infty}((x,v),\cdot)$

Sample $\tilde{h} \sim \mathrm{Unif}[0,\tilde{\ell}]$.
if $x - v\tilde{h} \notin [0,1]$ **then**
    Set $(x',v') = (x,-v)$.
else
    Set $(x',v') = (x - v\tilde{h}, v)$ with probability $m_{v,\infty}(x,\tilde{h})$, and $(x',v') = (x,-v)$ otherwise.
end if

The point of Theorem 2 is what it does *not* contain. We do not need to rescale time. The limit is a non-degenerate discrete-time Markov chain on $[0,1]$ in both the reversible and the non-reversible case. With $\ell_n \asymp n$ both samplers move $\mathcal{O}(1)$ distance per iteration, and the $\mathcal{O}(n^2)$-versus-$\mathcal{O}(n)$ distinction in [1] is absent.

Proposition 3. *Let π_∞ denote the* $\mathrm{Beta}(\alpha,\beta)$ *distribution. Then*

$$\left\| P^m_{\mathrm{R},\tilde{\ell},\infty}(x,\cdot) - \pi_\infty \right\|_{\mathrm{TV}} \longrightarrow 0 \qquad \textit{for } \pi_\infty\textit{-almost every } x,$$

and

$$\left\| P^m_{\mathrm{NR},\tilde{\ell},\infty}((x,v),\cdot) - \tilde{\pi}_\infty \right\|_{\mathrm{TV}} \longrightarrow 0 \qquad \textit{for } \tilde{\pi}_\infty\textit{-almost every } (x,v).$$

Proof. See Appendix B. □

8 Simulations

8.1 Efficiency Against Acceptance Rate

Theorem 2 allows us to compare the reversible and non-reversible samplers directly through their limiting kernels. We simulate $P_{\mathrm{R},\tilde{\ell},\infty}$ and $P_{\mathrm{NR},\tilde{\ell},\infty}$ using Algorithms 6 and 7 with the common target $\mathrm{Beta}(2,3)$. We consider 38 proposal scales. For each value of $\tilde{\ell}$, we

run $M = 5{,}000$ independent chains generated by each algorithm. Each chain is initialized from π_∞, so we do not need a burn-in period.

For each $\tilde{\ell}$, we estimate the effective sample size of the identity.

$$\frac{\widehat{\mathrm{ESS}}_X}{T} = \frac{\mathrm{Var}_{\pi_\infty}(X)}{\hat{\sigma}_X^2}, \qquad \mathrm{Var}_{\pi_\infty}(X) = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)},$$

where $\hat{\sigma}_X^2$ is estimated using the replicated lugsail batch-means estimator [38].

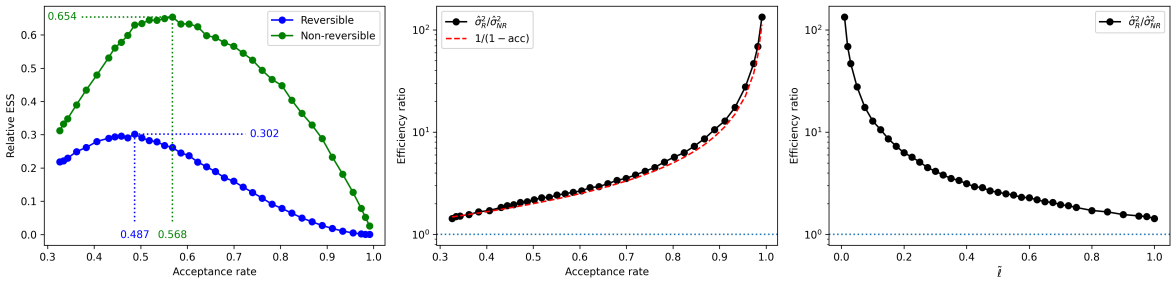


Figure 4: (Left) Estimated relative effective sample size for the observable $g(x) = x$ under the limiting kernels $P_{\mathrm{R},\tilde{\ell},\infty}$ (blue) and $P_{\mathrm{NR},\tilde{\ell},\infty}$ (green), plotted against the empirical acceptance rate. Both curves decrease towards zero as the acceptance rate approaches one; each has an interior maximum, and the non-reversible maximum occurs at a higher acceptance rate than the reversible maximum. (Middle) The efficiency ratio $\hat{\sigma}_{\mathrm{R}}^2/\hat{\sigma}_{\mathrm{NR}}^2$ plotted against the acceptance rate, together with the reference curve $1/(1 - \mathrm{acc})$ [32]. (Right) The efficiency ratio as a function of $\tilde{\ell}$.

After separately tuning the proposal scale, the non-reversible sampler is more than twice as efficient for this specific target for the component-proportion observable. The variance ratio $\hat{\sigma}_{\mathrm{R}}^2/\hat{\sigma}_{\mathrm{NR}}^2$ matches $1/(1 - \mathrm{acc})$ closely. This agrees well with the results of [32] and suggests that the speed-up from non-reversibility is most noticeable in the high-acceptance setting.

8.2 Informative Normal Mixture Experiment

We repeat the comparison for a two-component normal mixture model with observed data. We take $n = 1{,}000$, $\boldsymbol{\alpha} = (2, 3)$, observation variance $\sigma^2 = 1$, and independent component-mean priors

$$\theta_k \sim \mathcal{N}(0, \sigma_0^2), \qquad \sigma_0^2 = 0.005.$$

We generate a single dataset from this model and run our samplers for different values of ℓ. We choose a small σ_0^2 so that we can compare the samplers at acc ≈ 1. This choice creates greater uncertainty about each allocation.

We consider 38 proposal scales between 1 and 1,000. For each value of ℓ, we run $M = 5{,}000$ independent pairs of reversible and non-reversible chains. We simulate a long pilot run of length $T = 10^8$ at $\ell = 300$, discard a burn-in period of 10^6, and retain every 10,000th sample in a posterior bank so that we can initialize our simulations from approximate stationarity. Here, the efficiency observable is

$$g(c) = n_1(c).$$

For each sampler, the asymptotic variance is estimated with the same lugsail procedure as before.

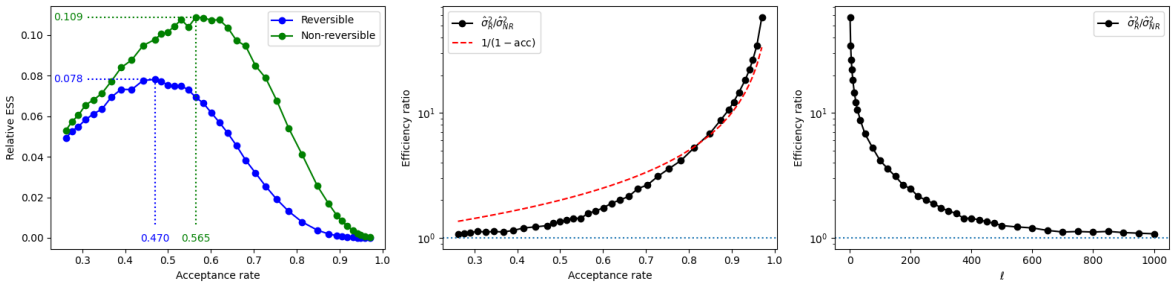


Figure 5: Efficiency comparison for the observable $g(c) = n_1(c)$ in the two-component normal mixture. (Left) Estimated relative effective sample size of $P_{\mathrm{R},\ell}$ (blue) and $P_{\mathrm{NR},\ell}$ (green) against empirical acceptance rate. (Middle) The efficiency ratio $\hat{\sigma}^2_{\mathrm{R}}/\hat{\sigma}^2_{\mathrm{NR}}$ plotted against the acceptance rate, together with the reference curve $1/(1 - \mathrm{acc})$ [32]. (Right) The efficiency ratio as a function of $\tilde{\ell}$.

As in the prior experiment, the qualitative behaviour is similar to what we expected, although the advantage of non-reversibility is smaller here.

8.3 Transient Phase

We next compare our algorithms in the transient phase. Ascolani and Zanella [1] observed a large advantage from non-reversibility in the transient phase when targeting the posterior from the normal mixture model described above. We consider a three-component normal mixture with

$$n = 1{,}000, \qquad \boldsymbol{\alpha} = (2, 1, 1), \qquad \mu_0 = 0, \qquad \sigma_0^2 = \sigma^2 = 1.$$

For each of the $M = 7{,}000$ independent repetitions, we first generate a dataset from the Bayesian model. That is, we draw the mixture weights and component means from their respective priors and then generate n observations from the resulting mixture. Conditional on each dataset, we run the reversible and non-reversible samplers from the same initialization,

$$c_i^{(0)} \overset{\text{i.i.d.}}{\sim} \mathrm{Unif}([K]).$$

We perform this experiment for proposal scales $\ell = 1$ and $\ell = 300$, using the same datasets and hence the same posteriors and initial allocations for both proposal scales. This diagnostic is based on the argument of Geweke [11]. If the posterior sampler has reached stationarity for each generated dataset, averaging its output over the repetitions recovers the prior distribution. Thus, agreement between the empirical distribution at a given time and its prior marginal distribution—which, under $\boldsymbol{\alpha} = (2, 1, 1)$, is $n_1 \sim \text{BetaBinomial}(n, 2, 2)$—indicates that the chain typically reaches stationarity by that time. We record the chains on two different time scales. For $\ell = 300$, the proposal size satisfies

$$h \sim \text{Unif}\{1, \ldots, 300\}, \qquad \mathbb{E}[h] = 150.5,$$

whereas $h = 1$ when $\ell = 1$. So we track the $\ell = 300$ chains at iterations

$$0, 1, 2, 3, 4, 5, 10, 25, 50,$$

and the $\ell = 1$ chains at 150 times these values.

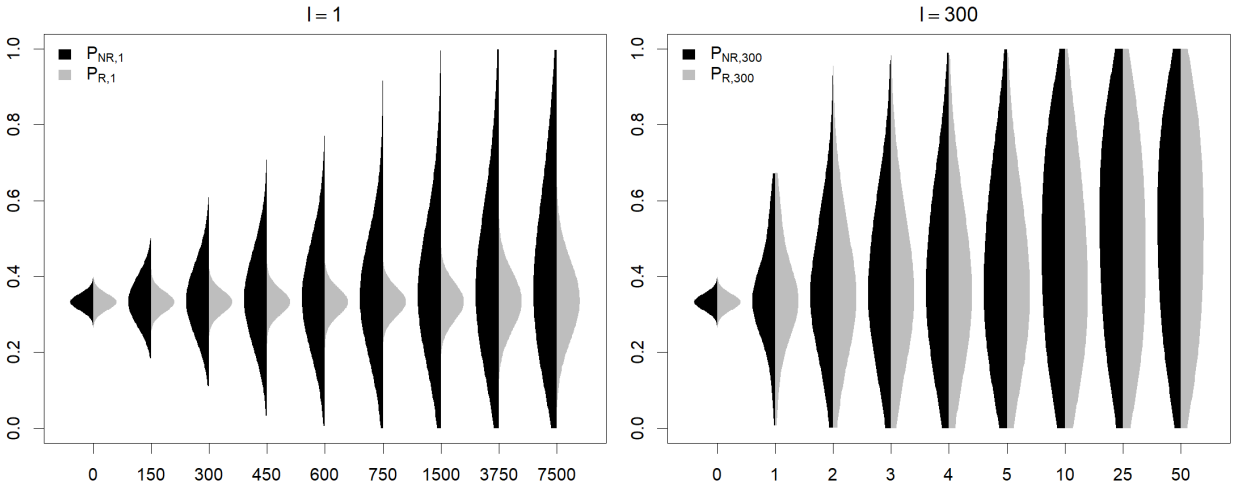


Figure 6: Transient distribution of the first component count $n_1(c)$ over $M = 7{,}000$ independent datasets. The left panel shows $P_{\text{R},1}$ (gray) and $P_{\text{NR},1}$ (black). The right panel shows $P_{\text{R},300}$ and $P_{\text{NR},300}$. Each split violin shows the empirical marginal distribution at the indicated number of Markov chain iterations.

Figure 6 shows that, with $\ell = 1$, the non-reversible chain converges to stationarity much more quickly than the reversible sampler. This closely matches the results in [1], in which the authors performed a similar experiment comparing the one-step P_{NR} and P_{MG}. Even after 7,500 iterations, $P_{\text{R},1}$ is still concentrated around its initial proportion, while $P_{\text{NR},1}$ is much closer to stationarity.

With $\ell = 300$, both $P_{\mathrm{R},300}$ and $P_{\mathrm{NR},300}$ reach stationarity very quickly, and the non-reversible sampler appears to converge only slightly faster. Moreover, under this time normalization, both multiple-allocation samplers approach the stationary distribution much more rapidly than $P_{\mathrm{NR},1}$. Thus, although non-reversibility gives a substantial advantage when proposals are microscopic, most of this advantage disappears once the proposal scale is made macroscopic.

Finally, we timed the simulations of the chains for this experiment and found that the $\ell = 1$ chains took 18.14 seconds, whereas the $\ell = 300$ chains took only 2.22 seconds. If we run both sets of chains for 7,500 iterations, which is far more than enough to reach stationarity with $\ell = 300$, the $\ell = 1$ chains take 23.64 seconds and the $\ell = 300$ chains take 290.25 seconds. Thus, the macroscopic chains appear to require fewer than 1/150 as many iterations to reach stationarity, while a 7,500-iteration macroscopic run is approximately 12.27 times as computationally expensive as a microscopic run of the same length. For this example, the advantage of scaling the proposal is apparent and well supported even when taking computational cost into account.

9 Conclusion

In this report, we introduced tunable multiple-allocation versions of two samplers from [1]: the reversible Metropolis–Hastings sampler and its non-reversible lifted counterpart. We proved invariance and ergodicity of the resulting kernels. For $K = 2$ in the prior case with $\alpha, \beta \geq 1$, we proved that when $\ell_n = \tilde{\ell} n$, both the reversible and non-reversible proportion chains converge weakly to non-degenerate discrete-time Markov chains without any time rescaling. Thus, the $\mathcal{O}(n^2)$-versus-$\mathcal{O}(n)$ comparison observed for chains with microscopic moves is absent. Our simulations support our theory even when the data are informative. Non-reversibility remains beneficial after tuning both algorithms, but the speed-up is moderate at best. The more substantial improvement comes from properly tuning either the reversible or the non-reversible algorithm.

In the future, we would like to extend Theorem 2 to $K > 2$ components. The required proof would not be substantially longer than the present one, but it would involve technical details concerning how to keep points strictly within the simplex and how to prove uniform convergence of the Metropolis acceptance probability. Extending our theory to cover $\alpha < 1$ or $\beta < 1$ is also desirable, but we suspect that different techniques may be required in that case.

A particularly important theoretical extension is the informative-posterior case. Ascolani and Zanella [1] note that the analysis in this case is likely to require measure-valued processes,

together with posterior asymptotic results for mixture models such as those in [13,24]. We are interested in determining whether a microscopic, macroscopic, or perhaps an intermediate $\ell = \mathcal{O}(n^{\frac{1}{2}})$ proposal scale is optimal when sampling from informative posteriors. We suspect that in the non-reversible case, under optimal scaling, the velocity variable may update far less frequently than the allocation variables. For this reason, multiscale methods such as those in [25] may be useful.

References

[1] Filippo Ascolani and Giacomo Zanella. A fast non-reversible sampler for Bayesian finite mixture models, 2025. arXiv:2510.03226.

[2] Joris Bierkens. Non-reversible Metropolis–Hastings. *Statistics and Computing*, 26(6):1213–1228, 2016. doi:10.1007/s11222-015-9598-x.

[3] Joris Bierkens, Paul Fearnhead, and Gareth O. Roberts. The Zig-Zag process and super-efficient sampling for Bayesian analysis of big data. *The Annals of Statistics*, 47(3):1288–1320, 2019. doi:10.1214/18-AOS1715.

[4] Alexandre Bouchard-Côté, Sebastian J. Vollmer, and Arnaud Doucet. The bouncy particle sampler: A nonreversible rejection-free Markov chain Monte Carlo method. *Journal of the American Statistical Association*, 113(522):855–867, 2018. doi:10.1080/01621459.2017.1294075.

[5] Gilles Celeux, Merrilee Hurn, and Christian P. Robert. Computational and inferential difficulties with mixture posterior distributions. *Journal of the American Statistical Association*, 95(451):957–970, 2000. doi:10.1080/01621459.2000.10474285.

[6] Fang Chen, László Lovász, and Igor Pak. Lifting Markov chains to speed up mixing. In *Proceedings of the Thirty-First Annual ACM Symposium on Theory of Computing*, pages 275–281. Association for Computing Machinery, 1999. doi:10.1145/301250.301315.

[7] Arthur P. Dempster, Nan M. Laird, and Donald B. Rubin. Maximum likelihood from incomplete data via the EM algorithm. *Journal of the Royal Statistical Society: Series B (Methodological)*, 39(1):1–22, 1977. doi:10.1111/j.2517-6161.1977.tb01600.x.

[8] Persi Diaconis, Susan Holmes, and Radford M. Neal. Analysis of a nonreversible Markov chain sampler. *The Annals of Applied Probability*, 10(3):726–752, 2000. doi:10.1214/aoap/1019487508.

[9] Jean Diebolt and Christian P. Robert. Estimation of finite mixture distributions through Bayesian sampling. *Journal of the Royal Statistical Society: Series B (Methodological)*, 56(2):363–375, 1994. doi:10.1111/j.2517-6161.1994.tb01985.x.

[10] Philippe Gagnon, Mylène Bèdard, and Alain Desgagné. An automatic robust Bayesian approach to principal component regression. *Journal of Applied Statistics*, 48(1):84–104, 2021. doi:10.1080/02664763.2019.1710478.

[11] John Geweke. Getting it right: Joint distribution tests of posterior simulators. *Journal of the American Statistical Association*, 99(467):799–804, 2004. doi:10.1198/016214504000001132.

[12] Bettina Grün and Gertraud Malsiner-Walli. Bayesian finite mixture models. In *Wiley StatsRef: Statistics Reference Online*, pages 1–19. John Wiley & Sons, 2022. doi:10.1002/9781118445112.stat08373.

[13] Aritra Guha, Nhat Ho, and XuanLong Nguyen. On posterior contraction of parameters and interpretability in Bayesian mixture modeling. *Bernoulli*, 27(4):2159–2188, 2021. doi:10.3150/20-BEJ1275.

[14] Paul Gustafson. A guided walk Metropolis algorithm. *Statistics and Computing*, 8(4):357–364, 1998. doi:10.1023/A:1008880707168.

[15] W. K. Hastings. Monte Carlo sampling methods using Markov chains and their applications. *Biometrika*, 57(1):97–109, 1970. doi:10.1093/biomet/57.1.97.

[16] James P. Hobert, Vivekananda Roy, and Christian P. Robert. Improving the convergence properties of the data augmentation algorithm with an application to Bayesian mixture modeling. *Statistical Science*, 26(3):332–351, 2011. doi:10.1214/11-STS365.

[17] Chii-Ruey Hwang, Shu-Yin Hwang-Ma, and Shuenn-Jyi Sheu. Accelerating Gaussian diffusions. *The Annals of Applied Probability*, 3(3):897–913, 1993. doi:10.1214/aoap/1177005371.

[18] Alan F. Karr. Weak convergence of a sequence of Markov chains. *Zeitschrift für Wahrscheinlichkeitstheorie und Verwandte Gebiete*, 33(1):41–48, 1975. doi:10.1007/BF00539859.

[19] David A. Levin and Yuval Peres. *Markov Chains and Mixing Times*. American Mathematical Society, Providence, RI, 2nd edition, 2017. doi:10.1090/mbk/107.

[20] Jean-Michel Marin, Kerrie L. Mengersen, and Christian P. Robert. Bayesian modelling and inference on mixtures of distributions. In Dipak K. Dey and C. R. Rao, editors, *Bayesian Thinking: Modeling and Computation*, volume 25 of *Handbook of Statistics*, pages 459–507. Elsevier, Amsterdam, 2005. doi:10.1016/S0169-7161(05)25016-2.

[21] Enzo Marinari and Giorgio Parisi. Simulated tempering: A new Monte Carlo scheme. *Europhysics Letters*, 19(6):451–458, 1992. doi:10.1209/0295-5075/19/6/002.

[22] Geoffrey J. McLachlan and David Peel. *Finite Mixture Models*. John Wiley & Sons, New York, 2000. doi:10.1002/0471721182.

[23] Nicholas Metropolis, Arianna W. Rosenbluth, Marshall N. Rosenbluth, Augusta H. Teller, and Edward Teller. Equation of state calculations by fast computing machines. *The Journal of Chemical Physics*, 21(6):1087–1092, 1953. doi:10.1063/1.1699114.

[24] XuanLong Nguyen. Convergence of latent mixing measures in finite and infinite mixture models. *The Annals of Statistics*, 41(1):370–400, 2013. doi:10.1214/12-AOS1065.

[25] Grigorios A. Pavliotis and Andrew M. Stuart. *Multiscale Methods: Averaging and Homogenization*, volume 53 of *Texts in Applied Mathematics*. Springer, New York, 2008. doi:10.1007/978-0-387-73829-1.

[26] Gareth O. Roberts. Optimal Metropolis algorithms for product measures on the vertices of a hypercube. *Stochastics and Stochastic Reports*, 62(3–4):275–283, 1998. doi:10.1080/17442509808834136.

[27] Gareth O. Roberts, Andrew Gelman, and Walter R. Gilks. Weak convergence and optimal scaling of random walk Metropolis algorithms. *The Annals of Applied Probability*, 7(1):110–120, 1997. doi:10.1214/aoap/1034625254.

[28] Gareth O. Roberts and Jeffrey S. Rosenthal. Markov chains and de-initializing processes. *Scandinavian Journal of Statistics*, 28(3):489–504, 2001. doi:10.1111/1467-9469.00250.

[29] Gareth O. Roberts and Jeffrey S. Rosenthal. Optimal scaling for various Metropolis–Hastings algorithms. *Statistical Science*, 16(4):351–367, 2001. doi:10.1214/ss/1015346320.

[30] Gareth O. Roberts and Jeffrey S. Rosenthal. General state space Markov chains and MCMC algorithms. *Probability Surveys*, 1:20–71, 2004. doi:10.1214/154957804100000024.

[31] Gareth O. Roberts and Jeffrey S. Rosenthal. Minimising MCMC variance via diffusion limits, with an application to simulated tempering. *The Annals of Applied Probability*, 24(1):131–149, 2014. doi:10.1214/12-AAP918.

[32] Gareth O. Roberts and Jeffrey S. Rosenthal. Quantifying the speed-up from non-reversibility in MCMC tempering algorithms, 2025. arXiv:2501.16506.

[33] Sebastian M. Schmon and Philippe Gagnon. Optimal scaling of random walk Metropolis algorithms using Bayesian large-sample asymptotics. *Statistics and Computing*, 32(2):28, 2022. doi:10.1007/s11222-022-10080-8.

[34] Jiangwei Shang, Yi-Lin Seah, Hui Khoon Ng, David John Nott, and Berthold-Georg Englert. Monte Carlo sampling from the quantum state space. I. *New Journal of Physics*, 17:043017, 2015. doi:10.1088/1367-2630/17/4/043017.

[35] Luke Tierney. Markov chains for exploring posterior distributions. *The Annals of Statistics*, 22(4):1701–1728, 1994. doi:10.1214/aos/1176325750.

[36] Francesco G. Tricomi and Arthur Erdélyi. The asymptotic expansion of a ratio of Gamma functions. *Pacific Journal of Mathematics*, 1(1):133–142, 1951. doi:10.2140/pjm.1951.1.133.

[37] Konstantin S. Turitsyn, Michael Chertkov, and Marija Vucelja. Irreversible Monte Carlo algorithms for efficient sampling. *Physica D: Nonlinear Phenomena*, 240(4–5):410–414, 2011. doi:10.1016/j.physd.2010.10.003.

[38] Dootika Vats and Christina Knudson. Revisiting the Gelman–Rubin diagnostic. *Statistical Science*, 36(4):518–529, 2021. doi:10.1214/20-STS812.

[39] Roman Vershynin. *High-Dimensional Probability: An Introduction with Applications in Data Science*. Cambridge Series in Statistical and Probabilistic Mathematics. Cambridge University Press, Cambridge, 2nd edition, 2026. doi:10.1017/9781009490672.

[40] Jun Yang, Gareth O. Roberts, and Jeffrey S. Rosenthal. Optimal scaling of random-walk Metropolis algorithms on general target distributions. *Stochastic Processes and their Applications*, 130(10):6094–6132, 2020. doi:10.1016/j.spa.2020.05.004.

A Invariance and Ergodicity of $P_{\mathrm{R},\ell}$ and $P_{\mathrm{NR},\ell}$

Attribution. Section A.1 is not our work. The following general framework for lifted Markov chains on finite state spaces, as well as Proposition 4 and its proof, are reproduced (with minor revisions) from [1]. Our application to the multiple-allocation kernels begins in Section A.2.

A.1 General Framework

Let π be a probability distribution on a finite space $\mathcal{C}$. Let $D \in \mathbb{N}_{\geq 1}$, and let $(K_{d,+})_{d\in[D]}$ and $(K_{d,-})_{d\in[D]}$ be Markov transition kernels on $\mathcal{C}$ such that

$$\pi(c)K_{d,+}(c,c') = \pi(c')K_{d,-}(c',c), \qquad \text{for all } c \neq c' \text{ and all } d \in [D]. \tag{20}$$

Define the Markov transition kernel on $\mathcal{C}$ by

$$K_{\mathrm{R}}(c,c') = \sum_{d=1}^{D} p_c(d)K_d(c,c'), \tag{21}$$

where $K_d = (K_{d,+} + K_{d,-})/2$ and the p_c are weights with $\sum_{d=1}^{D} p_c(d) = 1$ for all $c \in \mathcal{C}$ and

$$p_c(d) = p_{c'}(d) \qquad \text{if } K_d(c,c') > 0. \tag{22}$$

Define the Markov transition kernel on $\mathcal{C} \times \{-1,1\}^D$ by

$$K_{\mathrm{NR}}\big((c,v),(c',v')\big) = \sum_{d=1}^{D} p_c(d)\big(R_d K_{\mathrm{lift},d} R_d\big)\big((c,v),(c',v')\big), \tag{23}$$

where R_d is the flipping (refreshment) operator

$$R_d\big((c,v),(c',v')\big) = \mathbb{1}(c=c')\left[(1-\lambda)\mathbb{1}(v=v') + \lambda\,\mathbb{1}(v_{-d}=v'_{-d},\ v'_d=-v_d)\right] \tag{24}$$

for some fixed $\lambda \in [0,1]$, and

$$K_{\mathrm{lift},d}((c,v),(c',v')) = K_{d,v_d}(c,c')\,Q_{d,c,c'}(v,v') \tag{25}$$

with

$$Q_{d,c,c'}(v,v') = \mathbb{1}(v_{-d}=v'_{-d})\left[\mathbb{1}(c\neq c')\mathbb{1}(v_d=v'_d) + \mathbb{1}(c=c')\mathbb{1}(v_d=-v'_d)\right]. \tag{26}$$

Proposition 4. *Under* (20)–(26)*, K_{R} is π-reversible, and hence π-invariant, and K_{NR} is $\tilde{\pi}$-invariant with $\tilde{\pi}(c,v) = \pi(c)2^{-D}$.*

Proof. Consider first K_{R}. By (20), for every $c \neq c'$,

$$\begin{aligned} 2\pi(c)K_d(c,c') &= \pi(c)K_{d,+}(c,c') + \pi(c)K_{d,-}(c,c') \\ &= \pi(c')K_{d,-}(c',c) + \pi(c')K_{d,+}(c',c) = 2\pi(c')K_d(c',c). \end{aligned}$$

Thus, by (21) and (22),

$$\pi(c)K_{\mathrm{R}}(c,c') = \sum_{d=1}^{D} p_c(d)\pi(c)K_d(c,c') = \sum_{d=1}^{D} p_{c'}(d)\pi(c')K_d(c',c) = \pi(c')K_{\mathrm{R}}(c',c),$$

meaning that K_{R} is π-reversible.

For K_{NR}, fix $(c',v') \in \mathcal{C} \times \{-1,+1\}^D$. If $v = v'$ then $Q_{d,c,c'}(v,v') = \mathbb{1}(c \neq c')$ by (26), and (20) gives

$$\begin{aligned} \sum_{c\in\mathcal{C}} \tilde{\pi}(c,v)K_{\mathrm{lift},d}((c,v),(c',v')) &= \sum_{c\in\mathcal{C}} \pi(c)K_{d,v_d}(c,c')Q_{d,c,c'}(v,v')2^{-D} \\ &= \sum_{c\neq c'} \pi(c)K_{d,v'_d}(c,c')2^{-D} \\ &= \sum_{c\neq c'} \pi(c')K_{d,-v'_d}(c',c)2^{-D} = \tilde{\pi}(c',v')\left[1 - K_{d,-v'_d}(c',c')\right]. \end{aligned}$$

Similarly, if $v_{-d} = v'_{-d}$ and $v_d = -v'_d$, then $Q_{d,c,c'}(v,v') = \mathbb{1}(c = c')$ and

$$\begin{aligned} \sum_{c\in\mathcal{C}} \tilde{\pi}(c,v)K_{\mathrm{lift},d}((c,v),(c',v')) &= \sum_{c\in\mathcal{C}} \pi(c)K_{d,v_d}(c,c')Q_{d,c,c'}(v,v')2^{-D} \\ &= \pi(c')K_{d,-v'_d}(c',c')2^{-D} = \tilde{\pi}(c',v')K_{d,-v'_d}(c',c'). \end{aligned}$$

Summing the two expressions and using that $K_{\mathrm{lift},d}((c,v),(c',v')) = 0$ when $v_{-d} \neq v'_{-d}$,

$$\sum_{c,v} \tilde{\pi}(c,v)K_{\mathrm{lift},d}((c,v),(c',v')) = \tilde{\pi}(c',v'),$$

so $K_{\mathrm{lift},d}$ is $\tilde{\pi}$-invariant. Since R_d is trivially $\tilde{\pi}$-invariant, and a composition of invariant kernels is invariant, $R_dK_{\mathrm{lift},d}R_d$ is $\tilde{\pi}$-invariant. Finally, by (22),

$$\begin{aligned} \sum_{c,v} \tilde{\pi}(c,v)K_{\mathrm{NR}}((c,v),(c',v')) &= \sum_{d=1}^{D}\sum_{c,v} p_c(d)\tilde{\pi}(c,v)(R_dK_{\mathrm{lift},d}R_d)((c,v),(c',v')) \\ &= \sum_{d=1}^{D} p_{c'}(d) \sum_{c,v} \tilde{\pi}(c,v)(R_dK_{\mathrm{lift},d}R_d)((c,v),(c',v')) \\ &= \sum_{d=1}^{D} p_{c'}(d)\tilde{\pi}(c',v') = \tilde{\pi}(c',v'), \end{aligned}$$

and therefore K_{NR} is $\tilde{\pi}$-invariant. □

A.2 Verifying the Hypotheses for Our Kernels

We now check that $P_{\mathrm{R},\ell}$ and $P_{\mathrm{NR},\ell}$ satisfy (20)–(26). Here, $\mathcal{K}$ takes the place of $[D]$, so $D = K(K-1)/2$, and the refreshment rate is $\lambda = \xi/n$. Conditions (21) and (23)–(26) hold by construction. The two conditions requiring proof are the skew-detailed balance identity (20) and the weight condition (22).

Write $P_{(k,k'),+}$ and $P_{(k,k'),-}$ for the directional kernels: the positive kernel proposes moving $h \in [\ell]$ allocations from component k to component k', and the negative kernel proposes moving h allocations from k' to k.

Lemma 5 (Skew-detailed balance)**.** *For every $(k,k') \in \mathcal{K}$ and every $c \neq c'$ in $[K]^n$,*

$$\pi(c)P_{(k,k'),+}(c,c') = \pi(c')P_{(k,k'),-}(c',c).$$

Proof. If c and c' differ in more than ℓ allocations, then both sides vanish. Suppose $c' = c^{\mathcal{I}}$ for some $\mathcal{I}$ with $|\mathcal{I}| = h \in [\ell]$ and $c_i = k$ for all $i \in \mathcal{I}$, so that c' is obtained from c by moving those h allocations from k to k'.

For this transition, $P_{(k,k'),+}$ must draw the step size h and then select $\mathcal{I}$ from the $\binom{n_k(c)}{h}$ available subsets. The reverse transition is realized by $P_{(k,k'),-}$ from c', which must draw the same h and select $\mathcal{I}$ from the $\binom{n_{k'}(c')}{h} = \binom{n_{k'}(c)+h}{h}$ subsets of component k' in state c'. Hence, using (15),

$$\begin{aligned}
\pi(c)P_{(k,k'),+}(c,c') &= \pi(c)\frac{1}{\ell}\binom{n_k(c)}{h}^{-1}\min\left(1, \frac{\pi(c')\binom{n_{k'}(c)+h}{h}^{-1}}{\pi(c)\binom{n_k(c)}{h}^{-1}}\right) \\
&= \frac{1}{\ell}\min\left(\pi(c)\binom{n_k(c)}{h}^{-1},\ \pi(c')\binom{n_{k'}(c)+h}{h}^{-1}\right) \\
&= \pi(c')\frac{1}{\ell}\binom{n_{k'}(c)+h}{h}^{-1}\min\left(1, \frac{\pi(c)\binom{n_k(c)}{h}^{-1}}{\pi(c')\binom{n_{k'}(c)+h}{h}^{-1}}\right) \\
&= \pi(c')P_{(k,k'),-}(c',c).
\end{aligned}$$

□

Lemma 6 (Weight condition)**.** *If $c \neq c'$ and $P_{(k,k'),+}(c,c') + P_{(k,k'),-}(c,c') > 0$, then $p_c(k,k') = p_{c'}(k,k')$.*

Proof. The kernels $P_{(k,k'),\pm}$ move allocations only between components k and k', so positivity of either implies $n_k(c) + n_{k'}(c) = n_k(c') + n_{k'}(c')$. The claim follows from the definition (10) of p_c. □

By Lemmas 5 and 6, Proposition 4 applies, so $P_{\mathrm{R},\ell}$ is π-invariant and $P_{\mathrm{NR},\ell}$ is $\tilde{\pi}$-invariant.

Lemma 7 (Irreducibility and aperiodicity)**.** *Let $\pi(c) > 0$ for all $c \in [K]^n$, let $\ell \in [n]$, and let $\xi \in (0, n)$. Then $P_{\mathrm{R},\ell}$ and $P_{\mathrm{NR},\ell}$ are irreducible and aperiodic.*

Proof. Every feasible proposal may be accepted with positive probability, because $\pi > 0$ everywhere.

Irreducibility of $P_{\mathrm{R},\ell}$. Fix $c \neq c'$ and an index i with $c_i = k \neq k' = c'_i$. The pair (k, k') has $p_c(k, k') > 0$ since $n_k(c) \geq 1$, the direction $(k_-, k_+) = (k, k')$ has probability $1/2$, the step size $h = 1$ has probability $1/\ell$, and the index i is then selected and moved with positive probability. So c can be brought one allocation closer to c' with positive probability, and at most n such steps are needed to reach c'.

Irreducibility of $P_{\mathrm{NR},\ell}$. Because $\xi > 0$, a single allocation can be moved in either direction between any pair with $p_c(k, k') > 0$, exactly as for $P_{\mathrm{R},\ell}$. Starting from any (c, v), the chain can flip the velocity component $v_{k,k'}$ in two iterations, provided that $p_c(k, k') > 0$. If $p_c(k, k') = 0$, the chain can temporarily move an allocation into component k or k' from a third component k'', flip $v_{k,k'}$ in two iterations, and then return the allocation to k'' while leaving both $v_{k,k''}$ and $v_{k',k''}$ unchanged.

Aperiodicity. Aperiodicity of both $P_{\mathrm{NR},\ell}$ and $P_{\mathrm{R},\ell}$ follows from irreducibility and the existence of a state with period 1. Any state near the boundary has period 1; specifically, this holds whenever $n_k(c) < \ell$ for some k. From such a state, either algorithm can propose moving $h > n_k(c)$ allocations, and the proposal is rejected. Thus, for c near the boundary, $P_{\mathrm{R},\ell}(c, c) > 0$ and $P_{\mathrm{NR},\ell}((c, v), (c, v^{(\mathrm{flip})})) > 0$. Because $\xi > 0$, we also have $P_{\mathrm{NR},\ell}((c, v), (c, v)) > 0$. Therefore, $P_{\mathrm{NR},\ell}$ and $P_{\mathrm{R},\ell}$ are uniformly ergodic [19, Theorem 4.9]. □

B Invariance and Ergodicity of $P_{\mathrm{R},\tilde{\ell},\infty}$ and $P_{\mathrm{NR},\tilde{\ell},\infty}$

Proof of Proposition 3. For $v \in \{-1, +1\}$, define the proposal density

$$q_v(x, x') := \frac{1}{\tilde{\ell}} \mathbb{1}\{0 \leq v(x - x') \leq \tilde{\ell}\}$$

and the accepted-move density

$$k_v(x, x') := q_v(x, x') \min\left\{1, \frac{\pi_\infty(x')}{\pi_\infty(x)}\right\}.$$

The endpoint conventions of Section 7 affect only a Lebesgue-null set. Since $q_v(x, x') = q_{-v}(x', x)$, the usual Metropolis–Hastings calculation gives the skew-balance identity

$$\pi_\infty(x) k_v(x, x') = \pi_\infty(x') k_{-v}(x', x) \qquad \text{for almost every } (x, x'). \tag{27}$$

Let

$$\rho_v(x) := 1 - \int_0^1 k_v(x, y)\mathrm{d}y$$

be the probability that the lifted chain rejects and flips its velocity. Fix (x', v'). Under $\tilde{\pi}_\infty(\mathrm{d}x, \{v\}) = \frac{1}{2}\pi_\infty(x)\mathrm{d}x$, the density of incoming accepted-move mass at (x', v') is

$$\begin{aligned}\frac{1}{2}\int_0^1 \pi_\infty(x)k_{v'}(x, x')\mathrm{d}x &= \frac{1}{2}\pi_\infty(x')\int_0^1 k_{-v'}(x', x)\mathrm{d}x \\ &= \frac{1}{2}\pi_\infty(x')\big[1 - \rho_{-v'}(x')\big].\end{aligned}$$

The rejection contribution from $(x', -v')$ has density

$$\tfrac{1}{2}\pi_\infty(x')\rho_{-v'}(x').$$

The two terms sum to $\frac{1}{2}\pi_\infty(x')$, so $\tilde{\pi}_\infty$ is invariant for $P_{\mathrm{NR},\tilde{\ell},\infty}$. π-invariance for the reversible kernel follows from the standard Metropolis–Hastings proof [30].

π-irreducibility and aperiodicity of $P_{\mathrm{R},\tilde{\ell},\infty}$ follow from [35, Section 2.3.2]. The reversible proposal is symmetric with density strictly positive in a neighbourhood of 0, and the positive density region of the Beta density is the open and connected set $(0, 1)$.

For $P_{\mathrm{NR},\tilde{\ell},\infty}$, let $A \subseteq [0, 1] \times \{-1, +1\}$ satisfy $\tilde{\pi}_\infty(A) > 0$. Then, for at least one velocity $v \in \{-1, +1\}$, the corresponding slice $A_v := \{x \in [0, 1] : (x, v) \in A\}$ has positive π_∞-measure, and hence positive Lebesgue measure. Choose a sufficiently small interval $I \subsetneq (0, 1)$ such that $A_v \cap I$ has positive Lebesgue measure. From any starting state $(x, \pm)$, the chain can make finitely many accepted moves of length less than $\tilde{\ell}$ and flip velocity using an out-of-bounds proposal near a boundary when necessary, until it reaches a point on the appropriate side of I with velocity v. From there, given I is small enough, the next transition has a positive density throughout I. Since $A_v \cap I$ has positive Lebesgue measure, this transition lands in $A_v \cap I$ with positive probability. Thus, $P_{\mathrm{NR},\tilde{\ell},\infty}$ is $\tilde{\pi}_\infty$-irreducible. Because π_∞ is positive on $(0, 1)$, from any $(x, v) \notin A$ there exists a positive-probability path to A of length $m \geq 1$ involving no rejections except those triggered by an invalid proposal. Because the transition density is positive on $[0, \epsilon)$ for ϵ arbitrarily small, we can extend the path to $m+1$ time steps by splitting one of the steps in two. It follows that $P_{\mathrm{NR},\tilde{\ell},\infty}$ is aperiodic.

The total variation convergence for both algorithms then follows from [35, Theorem 1]. □

C Proof of Theorem 2

In this section, we prove that the chains on the proportion space generated by P_{R,ℓ_n} and P_{NR,ℓ_n} converge weakly, as $n \to \infty$, to the chains generated by the limiting kernels $P_{\mathrm{R},\tilde{\ell},\infty}$

and $P_{\mathrm{NR},\tilde{\ell},\infty}$ of Algorithms 6 and 7.

The argument is organized as follows. Section C.1 states the hypotheses of Karr's [18] theorem and sufficient conditions ensuring that our algorithms satisfy them. Section C.2 proves convergence results for the acceptance function $m_{v,n}$. The acceptance function is $m_{v,n}(x, \tilde{h}) = \min\{1, r_{v,n}(x, \tilde{h})\}$, where $r_{v,n}$ is the eight-Gamma ratio (18) derived in Section 7, and we show that $m_{v,n}$ converges *uniformly* to an MH acceptance function on an arbitrarily large compact subset of the $(x, \tilde{h})$-domain. Section C.3 shows that, for any limit point x, all but an arbitrarily small proportion of the admissible step sizes fall inside that compact set (Lemma 14). In Section C.4, we use all these results to complete the proof.

C.1 Setup, Standing Assumptions, and Notation

Theorem 8 (Karr [18], Theorem 4)**.** *Let E be a complete separable metric space with Borel σ-algebra $\mathcal{E}$, let $(\Omega, \mathcal{M}) := (E, \mathcal{E})^{\mathbb{N}}$ carry the product topology, and let $\mathbb{P}_j$ denote the law on $(\Omega, \mathcal{M})$ of the Markov chain with initial law ν_j and transition kernel P_j. If*

(a) $\nu_j \Rightarrow \nu$, and

(b) $P_j g(x_j) \to Pg(x)$ whenever $x_j \to x$ in E and $g : E \to \mathbb{R}$ is uniformly continuous and bounded,

then $\mathbb{P}_j \Rightarrow \mathbb{P}$, where $\mathbb{P}$ is the law of the chain with initial law ν and kernel P.

For the non-reversible chain, the state space is $E_{\mathrm{NR}} := [0, 1] \times \{-1, +1\}$, and for the reversible chain it is $E_{\mathrm{R}} := [0, 1]$. Both are compact metric spaces, so on either space a function $g : E \to \mathbb{R}$ is uniformly continuous if and only if it is continuous. Furthermore, any continuous g on a compact space is bounded. The function g may be written as a pair (g_+, g_-) with $g_v(x) := g(x, v)$.

Our proof will focus on the non-reversible kernel. The reversible-kernel limit then follows by taking $g_+ = g_- = g$ and averaging the $v = +1$ and $v = -1$ directional kernels without refreshment. The corresponding reversible result is Corollary 18.

Standing assumptions:

Assumption (A1). There are $K = 2$ components, and the Beta parameters satisfy $\alpha \geq 1$ and $\beta \geq 1$.

Assumption (A2). The base proposal scale is $\tilde{\ell} = a/b \in (0, 1) \cap \mathbb{Q}$, and the sequence of sample sizes is $n_j := jb$, $j \geq 1$. This ensures that $\ell_j := \tilde{\ell} n_j = ja \geq 1$ is a positive integer for all j.

Assumption (A3). The refreshment rate $\xi_j \equiv \xi < b$ is fixed and smaller than all n_j.

The realized step size $h \in [\ell_j]$ of Algorithms 4 and 5 is uniform on $\{1, \dots, \ell_j\}$. After rescaling by n_j it takes values in the grid

$$\mathcal{H}_{n_j} := \left\{ \tfrac{1}{n_j}, \tfrac{2}{n_j}, \dots, \tfrac{\ell_j}{n_j} = \tilde{\ell} \right\} \subset (0, \tilde{\ell}], \qquad |\mathcal{H}_{n_j}| = \ell_j = \tilde{\ell} n_j,$$

and we write $\tilde{h}$ for a generic element of $\mathcal{H}_{n_j}$.

Karr's theorem requires that all the kernels act on the same complete space. We therefore extend the discrete spaces to $[0,1]$ by projecting onto the nearest mesh point before step 1 of the algorithm,

$$\hat{x}_j := \frac{\lfloor n_j x_j + 1/2 \rfloor}{n_j}, \qquad \text{so that} \qquad |\hat{x}_j - x_j| \le \frac{1}{2n_j} \quad \text{and} \quad x_j \to x \implies \hat{x}_j \to x.$$

Let $\chi_v(x, \tilde{h}) := \mathbb{1}\{x - v\tilde{h} \in [0,1]\}$. This is the indicator on the set of valid proposals from x. For $n \in \mathbb{N} \cup \{\infty\}$, we write the general integrand as follows:

$$F_{v,n}(\tilde{h}; x) := \big(1 - \chi_v(x,\tilde{h})\big)\, g_{-v}(x) + \chi_v(x,\tilde{h}) \Big[m_{v,n}(x,\tilde{h})\, g_v(x - v\tilde{h}) + \big(1 - m_{v,n}(x,\tilde{h})\big) g_{-v}(x) \Big]. \tag{28}$$

Let P_j° be the extended finite kernel with the two refreshment steps removed. Then

$$P_j^\circ g(x_j, v) = \frac{1}{\ell_j} \sum_{\tilde{h} \in \mathcal{H}_{n_j}} F_{v,n_j}\big(\tilde{h}; \hat{x}_j\big), \qquad Pg(x,v) = \frac{1}{\tilde{\ell}} \int_0^{\tilde{\ell}} F_{v,\infty}(\tilde{h}; x) \mathrm{d}\tilde{h}. \tag{29}$$

Remark 9 (The refreshment error). Define the refreshment kernel by its action on g_v:

$$R_j g_v(x) := \left(1 - \frac{\xi}{n_j}\right) g_v(x) + \frac{\xi}{n_j} g_{-v}(x).$$

The true extended kernel is $P_j = R_j P_j^\circ R_j$. Since Markov kernels are contractions,

$$\begin{aligned}
\|P_j g - P_j^\circ g\|_\infty &= \|R_j P_j^\circ R_j g - R_j P_j^\circ g + R_j P_j^\circ g - P_j^\circ g\|_\infty \\
&\le \|R_j P_j^\circ (R_j g - g)\|_\infty + \|(R_j - I) P_j^\circ g\|_\infty \\
&\le \|R_j g - g\|_\infty + \|(R_j - I) P_j^\circ g\|_\infty \\
&\le \frac{2\xi}{n_j} \|g\|_\infty + \frac{2\xi}{n_j} \|P_j^\circ g\|_\infty \\
&\le \frac{4\xi}{n_j} \|g\|_\infty.
\end{aligned}$$

So we have $|P_j g_v(x) - P_j^\circ g_v(x)| \le \frac{4\xi}{n_j} \|g\|_\infty \longrightarrow 0$.

A convergent sequence $v_j \to v$ in the discrete set $\{-1, +1\}$ is eventually constant, so we may assume $v_j \equiv v$. Moreover, the map $(x, \alpha, \beta) \mapsto (1 - x, \beta, \alpha)$ carries the $v = +1$ dynamics onto the $v = -1$ dynamics, so it suffices to prove every statement below for the $v = +1$ case. We drop the subscript v from m and F and write $m_{n_j}(x, \tilde{h})$ and $F_{n_j}(\tilde{h}; x)$ for m_{+1,n_j} and F_{+1,n_j}.

Condition (a). To satisfy condition (a), we set $\nu_j \equiv \delta_{(a/b,+1)}$. We can choose any sequence $\nu_j \Rightarrow \nu$. What is important for interpreting our result is that we initialize $c^{(0)} = (c_1^{(0)}, \ldots, c_{n_j}^{(0)})$ such that $\mathcal{L}\left(c^{(0)} \mid X_0 = x\right) = \mathrm{Unif}\{c : X(c) = x\}$.

The error decomposition. Fix $x_j \to x$ and a continuous g. By Remark 9,

$$\left|P_j g_v(x_j) - P_j^\circ g_v(x_j)\right| \le \frac{4\xi}{n_j}\|g\|_\infty.$$

Hence, by the triangle inequality,

$$\begin{aligned}\left|P_j g_v(x_j) - P g_v(x)\right| &\le \left|P_j g_v(x_j) - P_j^\circ g_v(x_j)\right| \\ &\quad + \left|P_j^\circ g_v(x_j) - P g_v(x)\right| \\ &\le \frac{4\xi}{n_j}\|g\|_\infty + \left|P_j^\circ g_v(x_j) - P g_v(x)\right|.\end{aligned}$$

The first term tends to zero. We therefore focus on the remaining term, add and subtract

$$\frac{1}{\ell_j}\sum_{\tilde{h}\in\mathcal{H}_{n_j}} F_\infty(\tilde{h}; \hat{x}_j) \qquad \text{and} \qquad \frac{1}{\ell_j}\sum_{\tilde{h}\in\mathcal{H}_{n_j}} F_\infty(\tilde{h}; x).$$

Applying the triangle inequality to split the remaining term into three quantities:

$$\begin{aligned}\left|P_j^\circ g(x_j, v) - P g(x, v)\right| \le{}& \underbrace{\frac{1}{\ell_j}\sum_{\tilde{h}\in\mathcal{H}_{n_j}} \left|F_{n_j}(\tilde{h}; \hat{x}_j) - F_\infty(\tilde{h}; \hat{x}_j)\right|}_{A_j \text{ (acceptance error)}} \\ &+ \underbrace{\frac{1}{\ell_j}\sum_{\tilde{h}\in\mathcal{H}_{n_j}} \left|F_\infty(\tilde{h}; \hat{x}_j) - F_\infty(\tilde{h}; x)\right|}_{B_j \text{ (continuity error)}} \\ &+ \underbrace{\left|\frac{1}{\ell_j}\sum_{\tilde{h}\in\mathcal{H}_{n_j}} F_\infty(\tilde{h}; x) - \frac{1}{\tilde{\ell}}\int_0^{\tilde{\ell}} F_\infty(\tilde{h}; x)\mathrm{d}\tilde{h}\right|}_{C_j \text{ (Riemann-sum error)}}.\end{aligned}$$

To finish our proof, we will show that as, $j \to \infty$, each of A_j, B_j, and C_j converges to zero. Since (28) is a convex combination of values of g,

$$|F_n(\tilde{h}; x)| \leq \|g\|_\infty$$

for every n, $\tilde{h}$, and x. Consequently, each summand in A_j and B_j is bounded by $2\|g\|_\infty$. We will show that these summands converge to zero except on a set of step sizes whose relative size can be made arbitrarily small.

C.2 The Acceptance Function

Write $m_n(x, \tilde{h}) = \min\{1, r_n(x, \tilde{h})\}$ with r_n the acceptance ratio (18), and group the eight Γ's by setting

$$G_a(z) := \frac{\Gamma(z+a)}{\Gamma(z+1)}, \qquad z \geq 0,\ a \geq 1, \tag{30}$$

so that

$$r_n(x, \tilde{h}) = \underbrace{\frac{G_\alpha\big(n(x-\tilde{h})\big)}{G_\alpha(nx)}}_{=:\ \lambda_n \text{ (left factor)}} \times \underbrace{\frac{G_\beta\big(n(1-x+\tilde{h})\big)}{G_\beta\big(n(1-x)\big)}}_{=:\ \rho_n \text{ (right factor)}}. \tag{31}$$

We use three properties of G_a throughout: it is continuous and positive on all of $[0, \infty)$, with $G_a(0) = \Gamma(a)$; $G_1 \equiv 1$; and $G_a(nz) \sim (nz)^{a-1}$ as $n \to \infty$ for each fixed $z > 0$ [36].

The pointwise limit. The domain of $m_n(x, \tilde{h})$ is $\mathcal{D} := \{(x, \tilde{h}) : 0 \leq \tilde{h} \leq \min(x, \tilde{\ell})\}$.

Lemma 10 (Pointwise limit of m_n)**.** *Let $\pi_\infty(x) := x^{\alpha-1}(1-x)^{\beta-1}$. This is the unnormalized* Beta(α, β) *density. For every fixed $(x, \tilde{h}) \in \mathcal{D}$, the limit $m_\infty(x, \tilde{h}) := \lim_{n\to\infty} m_n(x, \tilde{h})$ exists. Wherever the density ratio is determinate,*

$$m_\infty(x, \tilde{h}) = \min\left\{1, \frac{\pi_\infty(x-\tilde{h})}{\pi_\infty(x)}\right\}. \tag{32}$$

The ratio is of the form $0/0$ only at $(0,0)$ when $\alpha > 1$ and at $(1,0)$ when $\beta > 1$. At those two points we define $m_\infty = 1$. This agrees with m_n for all n.

Proof. The ratio in (32) is $\left(\frac{x-\tilde{h}}{x}\right)^{\alpha-1}\left(\frac{1-x+\tilde{h}}{1-x}\right)^{\beta-1}$, which is $\lim_{n\to\infty} \lambda_n \cdot \rho_n$ whenever $0 < \tilde{h} < x < 1$. The remaining points of $\mathcal{D}$ are those at which an argument is frozen at 0:

- $\tilde{h} = 0$: then $\lambda_n \equiv \rho_n \equiv 1$, so $m_n \equiv 1$.

- $0 < \tilde{h} = x$ (hence $x \le \tilde{\ell} < 1$): $G_\alpha(0) = \Gamma(\alpha)$ is fixed while $G_\alpha(nx) \sim (nx)^{\alpha-1}$. Hence $\lambda_n \to 0$ if $\alpha > 1$, and $\lambda_n \equiv 1$ if $\alpha = 1$. On the right we have,

$$\rho_n = \frac{G_\beta(n)}{G_\beta(n(1-x))} \longrightarrow (1-x)^{1-\beta},$$

which is finite and positive since $x < 1$. Therefore

$$r_n(x,x) \longrightarrow \begin{cases} 0, & \alpha > 1, \\ (1-x)^{1-\beta}, & \alpha = 1, \end{cases} = \frac{\pi_\infty(0)}{\pi_\infty(x)}.$$

- If $x = 1$ and $\tilde{h} > 0$, then $G_\beta(0) = \Gamma(\beta)$ is fixed while $G_\beta(n\tilde{h}) \sim (n\tilde{h})^{\beta-1}$. Hence

$$r_n(1,\tilde{h}) \longrightarrow \begin{cases} +\infty, & \beta > 1, \\ (1-\tilde{h})^{\alpha-1}, & \beta = 1, \end{cases} = \frac{\pi_\infty(1-\tilde{h})}{\pi_\infty(1)}$$

since

$$\lambda_n = \frac{G_\alpha(n(1-\tilde{h}))}{G_\alpha(n)} \longrightarrow (1-\tilde{h})^{\alpha-1} > 0,$$

while $\rho_n \to \infty$ if $\beta > 1$ and $\rho_n \equiv 1$ if $\beta = 1$.

□

Thus, m_∞ is the Metropolis–Hastings acceptance probability for the Beta(α, β) density π_∞ under the $\tilde{h} \sim \text{Unif}[0, \tilde{\ell}]$ proposal. Moreover, m_∞ is continuous on $\mathcal{D} \setminus \{(0,0),(1,0)\}$.

Uniform limits. Propositions 15 and 16 require more than Lemma 10: the sums there are over $\tilde{h}$-grids whose points move with j, so we need uniform convergence. We prove uniform convergence on two regions chosen so that Gamma-ratio asymptotics apply separately to λ_n and ρ_n [36].

Lemma 11 (Uniform Convergence Gamma Ratios)**.** *Let $a, b > 0$ and let $\mathfrak{C} \subseteq [\eta_1, \zeta_1] \times [\eta_2, \zeta_2]$ be compact with $\eta_1, \eta_2 > 0$. Then*

$$\lim_{n\to\infty} \sup_{(z_1,z_2)\in\mathfrak{C}} \left| \frac{\Gamma(nz_1+a)}{\Gamma(nz_1+b)} \frac{\Gamma(nz_2+b)}{\Gamma(nz_2+a)} - \left(\frac{z_1}{z_2}\right)^{a-b} \right| = 0.$$

Proof. By [36], $\Gamma(w+a)/\Gamma(w+b) = w^{a-b}\big(1 + \mathcal{O}(w^{-1})\big)$ as $w \to \infty$. Applying this with $w = nz_1$ and $w = nz_2$,

$$\left| \frac{\Gamma(nz_1+a)}{\Gamma(nz_1+b)} \frac{\Gamma(nz_2+b)}{\Gamma(nz_2+a)} - \left(\frac{z_1}{z_2}\right)^{a-b} \right| = \left(\frac{z_1}{z_2}\right)^{a-b} \left(\mathcal{O}\left(\frac{1}{n\eta_1}\right) + \mathcal{O}\left(\frac{1}{n\eta_2}\right) + o\left(\frac{1}{n}\right) \right).$$

The prefactor $(z_1/z_2)^{a-b}$ is continuous on the compact set $\mathfrak{C}$ and is therefore bounded there by some $M < \infty$, and $z_i^{-1} \le \eta_i^{-1}$. So we have:

$$\sup_{(z_1,z_2)\in\mathfrak{C}} \left| \frac{\Gamma(nz_1+a)}{\Gamma(nz_1+b)} \frac{\Gamma(nz_2+b)}{\Gamma(nz_2+a)} - \left(\frac{z_1}{z_2}\right)^{a-b} \right| \le M\left(\mathcal{O}\left(\frac{1}{n\eta_1}\right) + \mathcal{O}\left(\frac{1}{n\eta_2}\right) + o\left(\frac{1}{n}\right)\right) \xrightarrow{n\to\infty} 0.$$

□

Lemma 12 (Interior region)**.** *For $\varepsilon \in (0,1)$ set*

$$\mathfrak{K}_\varepsilon := \left\{(x,\tilde{h}) \in [0,1]\times[0,\tilde{\ell}] \ : \ x - \tilde{h} \ge \varepsilon,\ 1-x \ge \varepsilon\right\}.$$

Then $m_n \to m_\infty$ uniformly on $\mathfrak{K}_\varepsilon$.

Proof. On $\mathfrak{K}_\varepsilon$, $x-\tilde{h}$, x, $1-x+\tilde{h}$, and $1-x$ are all bounded and bounded away from 0. It follows from Lemma 11 that with $(z_1,z_2) = (x-\tilde{h}, x)$ and $(a,b) = (\alpha, 1)$, λ_n converges uniformly. And with $(z_1,z_2) = (1-x+\tilde{h}, 1-x)$ and $(a,b) = (\beta,1)$, ρ_n converges uniformly. Both limits are bounded on $\mathfrak{K}_\varepsilon$, so their product r_n converges uniformly to the Beta(α,β) ratio in (32). Finally, $w \mapsto \min\{1,w\}$ is 1-Lipschitz, so $m_n \to m_\infty$. □

The wall $x = 1$ is not covered by the previous argument, since $1-x \to 0$ there and ρ_n blows up when $\beta > 1$. The blow-up is benign. It pushes r_n above 1, so acceptance is identically 1 in a boundary layer and no expansion is needed.

Lemma 13 (Wall region)**.** *Assume $\beta > 1$ and let $\eta \in (0, \tilde{\ell})$. Set $\sigma := \frac{1}{2}(1-\tilde{\ell}) > 0$. There exist $\delta = \delta(\eta,\alpha,\beta) \in (0, \min\{\sigma,\eta\})$ and n_0 such that*

$$m_n(x,\tilde{h}) \equiv 1 \equiv m_\infty(x,\tilde{h}) \qquad \text{for all } n \ge n_0 \text{ and all } (x,\tilde{h}) \in \mathfrak{W}_{\eta,\delta},$$

where $\mathfrak{W}_{\eta,\delta} := \{(x,\tilde{h}) : \eta \le \tilde{h} \le \tilde{\ell},\ 1-\delta \le x \le 1\}$.

Proof. First, G_a is increasing on $[0,\infty)$ for $a \ge 1$. This is because $G_a > 0$ and $\frac{\mathrm{d}}{\mathrm{d}z}\log G_a(z) = \psi(z+a) - \psi(z+1) \ge 0$, since the digamma function ψ is increasing on $(0,\infty)$.

Suppose $\beta > 1$ and let $(x,\tilde{h}) \in \mathfrak{W}_{\eta,\delta}$ with $\delta < \sigma$. Then

$$x - \tilde{h} \ge (1-\delta) - \tilde{\ell} \ge \sigma, \qquad x \le 1, \qquad 1 - x + \tilde{h} \ge \tilde{h} \ge \eta, \text{ and} \qquad 1 - x \le \delta.$$

Using the fact that G_α and G_β are increasing gives the deterministic bound

$$r_n(x,\tilde{h}) = \lambda_n \rho_n \ \ge \ \frac{G_\alpha(n\sigma)}{G_\alpha(n)} \frac{G_\beta(n\eta)}{G_\beta(n\delta)} \ =: \ \Lambda_n(\eta,\delta). \tag{33}$$

Note that the lower bound does not depend on $(x,\tilde{h})$. By [36], $\Lambda_n(\eta,\delta) \to \sigma^{\alpha-1}(\eta/\delta)^{\beta-1}$. Since $\beta > 1$ and σ is a fixed positive constant, the limit tends to $+\infty$ as $\delta \downarrow 0$ with η fixed. Thus, for sufficiently small δ and $n \ge n_0$, we have $r_n \ge 1$ and $m_n \equiv 1$ on $\mathfrak{W}_{\eta,\delta}$. The same bound applied to the ratio in (32) gives $m_\infty \equiv 1$ on $\mathfrak{W}_{\eta,\delta}$ as well. □

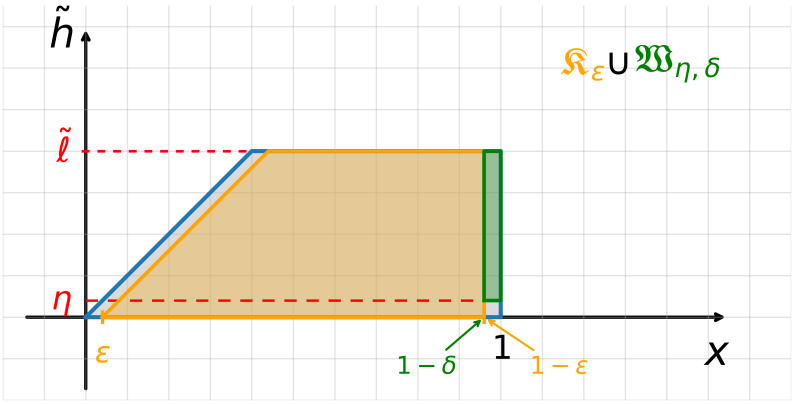


Figure 7: Uniform convergence is proved separately on the two regions. Both η and δ may be chosen arbitrarily small.

Note that if instead $\beta = 1$, then $\rho_n \equiv 1$, and the conclusion of Lemma 12 holds on $\{(x, \tilde{h}) : x - \tilde{h} \geq \varepsilon\}$ with no constraint on $1 - x$. So the following bounds for $\beta > 1$ are *more crude* than necessary when $\beta = 1$.

C.3 The Good Set of Step Sizes

Fix $\eta \in (0, \tilde{\ell})$ and let δ be as in Lemma 13. Define the *good set*

$$\mathfrak{G}_\eta := \mathfrak{K}_\delta \cup \mathfrak{W}_{\eta,\delta}. \tag{34}$$

The interior region is defined using δ and extends to $x = 1 - \delta$, where $\mathfrak{W}_{\eta,\delta}$ begins; thus, the two regions overlap without leaving a gap. Lemmas 12 and 13 give

$$\sup_{\mathfrak{G}_\eta} \left| m_n - m_\infty \right| \xrightarrow[n\to\infty]{} 0. \tag{35}$$

The complement $\mathcal{D} \setminus \mathfrak{G}_\eta$ is the strip along the diagonal $\tilde{h} = x$ together with the box at the corner $(1, 0)$ drawn in Figure 7. The proof in Section C.4 relies on the fact that for each fixed x, the complement is a $\tilde{h}$-range of length at most a constant multiple of η.

Lemma 14 (The bad range is short)**.** *For every $x \in [0, 1]$,*

$$\left| \{\tilde{h} \in [0, \tilde{\ell}] : (x, \tilde{h}) \in \mathcal{D} \setminus \mathfrak{G}_\eta\} \right| \;\leq\; 2\eta.$$

Proof. Fix x and take $(x, \tilde{h}) \in \mathcal{D} \setminus \mathfrak{G}_\eta$. Failing to lie in $\mathfrak{K}_\delta$ means $x - \tilde{h} < \delta$ or $1 - x < \delta$. The first puts $\tilde{h} \in (x - \delta, x]$, an interval of length δ. In the second case, $x > 1 - \delta$, so $(x, \tilde{h}) \in \mathfrak{W}_{\eta,\delta}$ unless $\tilde{h} < \eta$, i.e. $\tilde{h} \in [0, \eta)$. The bad range is contained in the union of these two intervals. Thus, the total length is at most $\delta + \eta \leq 2\eta$. □

Since $\mathcal{H}_{n_j}$ is a uniform grid with spacing $1/n_j$, an interval of length L contains at most Ln_j+1 of its points, so the number of grid step sizes with $(x,\tilde{h}) \in \mathcal{D}\setminus\mathfrak{G}_\eta$ is at most $2\eta n_j+2$. Thus, the corresponding upper bound on the proportion of the grid is

$$\frac{2\eta n_j+2}{\ell_j} = \frac{2\eta n_j+2}{\tilde{\ell} n_j} \xrightarrow[j\to\infty]{} \frac{2\eta}{\tilde{\ell}}. \tag{36}$$

For fixed x, $F_n(\cdot;x)$ is defined on all of $[0,\tilde{\ell}]$, whereas $m_n(x,\cdot)$ is defined only on the corresponding section of $\mathcal{D}$. For $\tilde{h} > x$, the indicator vanishes, $\chi_+(x,\tilde{h}) = 0$, and (28) collapses to

$$F_n(\tilde{h};x) = g_-(x) \qquad \text{for all } \tilde{h} > x \text{ and all } n \in \mathbb{N}\cup\{\infty\}. \tag{37}$$

C.4 Proof of Condition (b)

Proposition 15. $A_j \to 0$.

Proof. From (28), the two integrands differ only through the acceptance probability, so for any n and any $x \in [0,1]$:

$$\begin{aligned}
&F_n(\tilde{h};x) - F_\infty(\tilde{h};x)\\
&= \big(1-\chi_+(x,\tilde{h})\big)g_-(x) + \chi_+(x,\tilde{h})\Big[m_n(x,\tilde{h})g_+(x-\tilde{h}) + \big(1-m_n(x,\tilde{h})\big)g_-(x)\Big]\\
&-\big(1-\chi_+(x,\tilde{h})\big)g_-(x) - \chi_+(x,\tilde{h})\Big[m_\infty(x,\tilde{h})g_+(x-\tilde{h}) + \big(1-m_\infty(x,\tilde{h})\big)g_-(x)\Big].\\
&= \chi_+(x,\tilde{h})\big(m_n(x,\tilde{h}) - m_\infty(x,\tilde{h})\big)\big[g_+(x-\tilde{h}) - g_-(x)\big]
\end{aligned}$$

Hence

$$\Big|F_n(\tilde{h};x) - F_\infty(\tilde{h};x)\Big| \le 2\|g\|_\infty \Big|m_n(x,\tilde{h}) - m_\infty(x,\tilde{h})\Big|.$$

Therefore,

$$A_j \le \frac{2\|g\|_\infty}{\ell_j} \sum_{\tilde{h}\in\mathcal{H}_{n_j}} \Big|m_{n_j}(\hat{x}_j,\tilde{h}) - m_\infty(\hat{x}_j,\tilde{h})\Big|.$$

We partition into the three ranges:

- *Out of bounds:* $\tilde{h} > x$, on which (37) holds. Thus, $\Big|F_n(\tilde{h};x) - F_\infty(\tilde{h};x)\Big| = 0$.
- *The bad range* inside $\mathcal{D}$, of length $\le 2\eta$ by Lemma 14.
- *The good range* $\{\tilde{h} : (x,\tilde{h}) \in \mathfrak{G}_\eta\}$, on which (35) applies.

Hence,

$$A_j \le 2\|g\|_\infty \frac{2\eta n_j+2}{\ell_j} + 2\|g\|_\infty \sup_{\mathfrak{G}_\eta} |m_{n_j} - m_\infty|,$$

and by (36), $\limsup_{j\to\infty} A_j \le 4\|g\|_\infty \eta/\tilde{\ell}$. As $\eta > 0$ was arbitrary, $A_j \to 0$. □

Proposition 16 (Continuity error)**.** $B_j \to 0$.

Proof. Fix $\eta > 0$ and use the corresponding δ in the definition of $\mathfrak{G}_\eta$. Since g_- is uniformly continuous, define the modulus of continuity by

$$\omega_-(t) := \sup_{|y-z|\le t} |g_-(y) - g_-(z)|.$$

We have $\omega_-(t) \to 0$ as $t \downarrow 0$. The function F_∞ is continuous on the compact good set $\mathfrak{G}_\eta$. Hence, it is uniformly continuous there, and we define another modulus of continuity:

$$\omega_\eta(t) := \sup\left\{|F_\infty(\tilde{h}; y) - F_\infty(\tilde{h}; z)| : (y, \tilde{h}), (z, \tilde{h}) \in \mathfrak{G}_\eta,\ |y - z| \le t\right\}.$$

For sufficiently large j, we have $|\hat{x}_j - x| < \delta$. We split the step-size grid into three parts.

- *Both out of bounds:* If $\tilde{h} > \max\{x, \hat{x}_j\}$, then the proposal crosses the wall from both states. By (37), both terms evaluate g_- so $|F_\infty(\tilde{h}; \hat{x}_j) - F_\infty(\tilde{h}; x)| \le \omega_-(|\hat{x}_j - x|)$.
- *The common good range:* If both $(\hat{x}_j, \tilde{h})$ and $(x, \tilde{h})$ belong to $\mathfrak{G}_\eta$, then by the definition of ω_η, we have $|F_\infty(\tilde{h}; \hat{x}_j) - F_\infty(\tilde{h}; x)| \le \omega_\eta(|\hat{x}_j - x|)$.
- *The remaining grid points:* Every remaining point lies in the bad range associated with x or with $\hat{x}_j$. So we use our crude bound $|F_\infty(\tilde{h}; x)| \le \|g\|_\infty$ and the fact that all remaining points lie in the union of the bad ranges for x and $\hat{x}_j$. By Lemma 14, their union contains at most $4\eta n_j + 4$ grid points. And we have

$$|F_\infty(\tilde{h}; \hat{x}_j) - F_\infty(\tilde{h}; x)| \le 2\|g\|_\infty.$$

Summing over $\mathcal{H}_{n_j}$ gives

$$B_j \le \omega_-(|\hat{x}_j - x|) + \omega_\eta(|\hat{x}_j - x|) + 2\|g\|_\infty \frac{4\eta n_j + 4}{\ell_j}.$$

Since $|\hat{x}_j - x| \to 0$ and $\ell_j = \tilde{\ell} n_j$,

$$\limsup_{j\to\infty} B_j \le \frac{8\eta}{\tilde{\ell}} \|g\|_\infty.$$

Finally, letting $\eta \downarrow 0$ gives $B_j \to 0$. □

Proposition 17. $C_j \to 0$.

Proof. For fixed x, the function $\tilde{h} \mapsto F_\infty(\tilde{h}; x)$ is bounded by $\|g\|_\infty$ and, by Lemma 10, continuous on $[0, \tilde{\ell}]$ except possibly at the point $\tilde{h} = x$. A bounded function on a compact interval with finitely many discontinuities is Riemann integrable. Moreover, $\frac{1}{\ell_j} \sum_{\tilde{h} \in \mathcal{H}_{n_j}} F_\infty(\tilde{h}; x)$ is precisely the right-endpoint Riemann sum of $\frac{1}{\tilde{\ell}} \int_0^{\tilde{\ell}} F_\infty(\tilde{h}; x) \mathrm{d}\tilde{h}$ with mesh $1/n_j \to 0$. Hence, $C_j \to 0$. □

Corollary 18 (Reversible case)**.** *The same conclusion holds for P_{R,ℓ_j} and $P_{\mathrm{R},\tilde{\ell},\infty}$ on $E_{\mathrm{R}} = [0, 1]$.*

Proof. Algorithm 6 draws $v \sim \mathrm{Unif}\{-1, +1\}$ at each step and does not carry v forward, so

$$P_{\mathrm{R},\ell_j} g(x_j) = \tfrac{1}{2} \sum_{v \in \{-1,+1\}} \frac{1}{\ell_j} \sum_{\tilde{h} \in \mathcal{H}_{n_j}} F_{v,n_j}(\tilde{h}; \hat{x}_j),$$

which is (29) with $g_+ = g_- = g$ summed over both velocities and with no refreshment term. Each of the two summands converges to the corresponding limit by our non-reversible result (for $v = +1$) and by the symmetry $(x, \alpha, \beta) \mapsto (1 - x, \beta, \alpha)$ (for $v = -1$); hence, their sum converges □

Condition (a) holds by the choice of ν_j made in Section C.1. Remark 9, together with Propositions 15, 16, and 17, shows that every term in the error decomposition tends to zero. Hence, $P_j g(x_j, v) \to Pg(x, v)$ whenever $(x_j, v) \to (x, v)$ in E_{NR}. Theorem 8 therefore applies to P_{NR,ℓ_j} and $P_{\mathrm{NR},\tilde{\ell},\infty}$; by Corollary 18, it also applies to P_{R,ℓ_j} and $P_{\mathrm{R},\tilde{\ell},\infty}$. This proves Theorem 2.